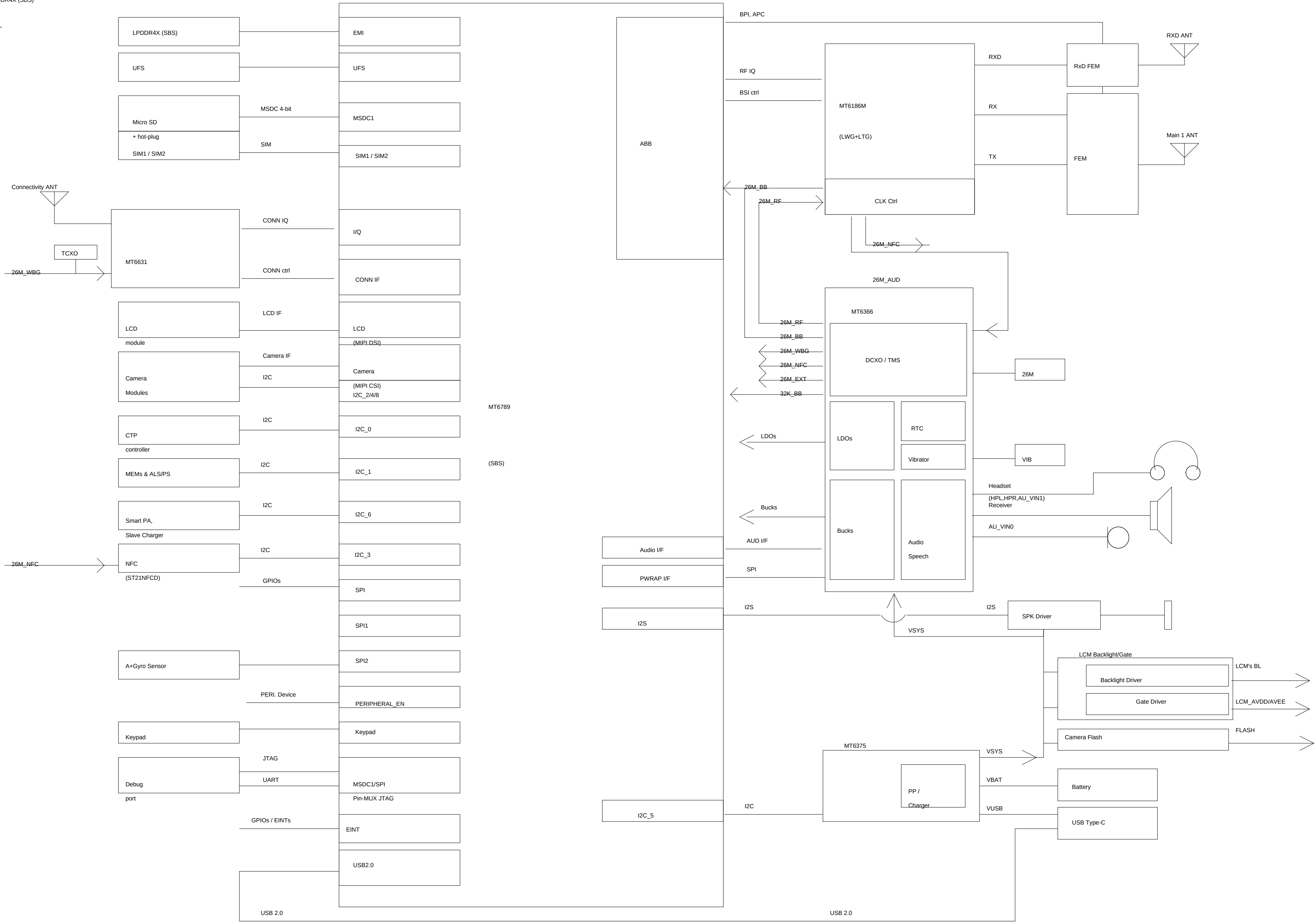


REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

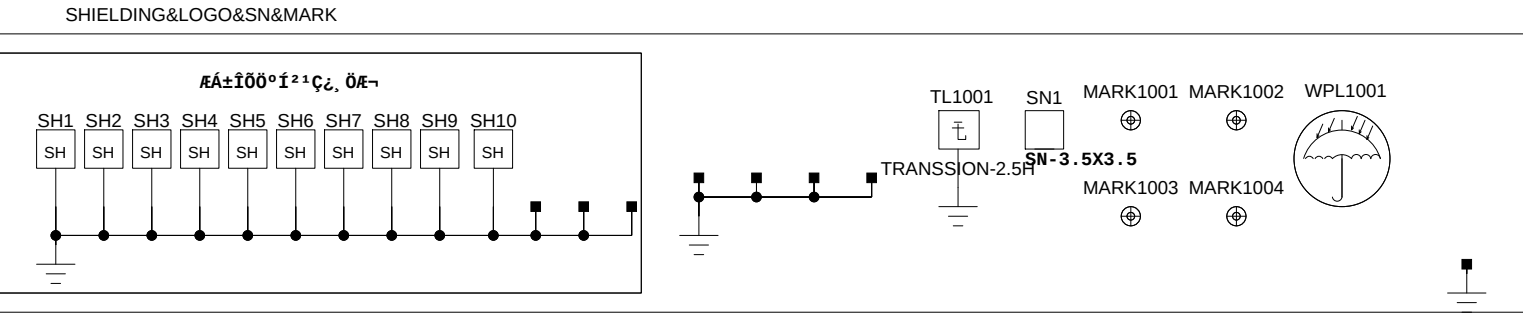
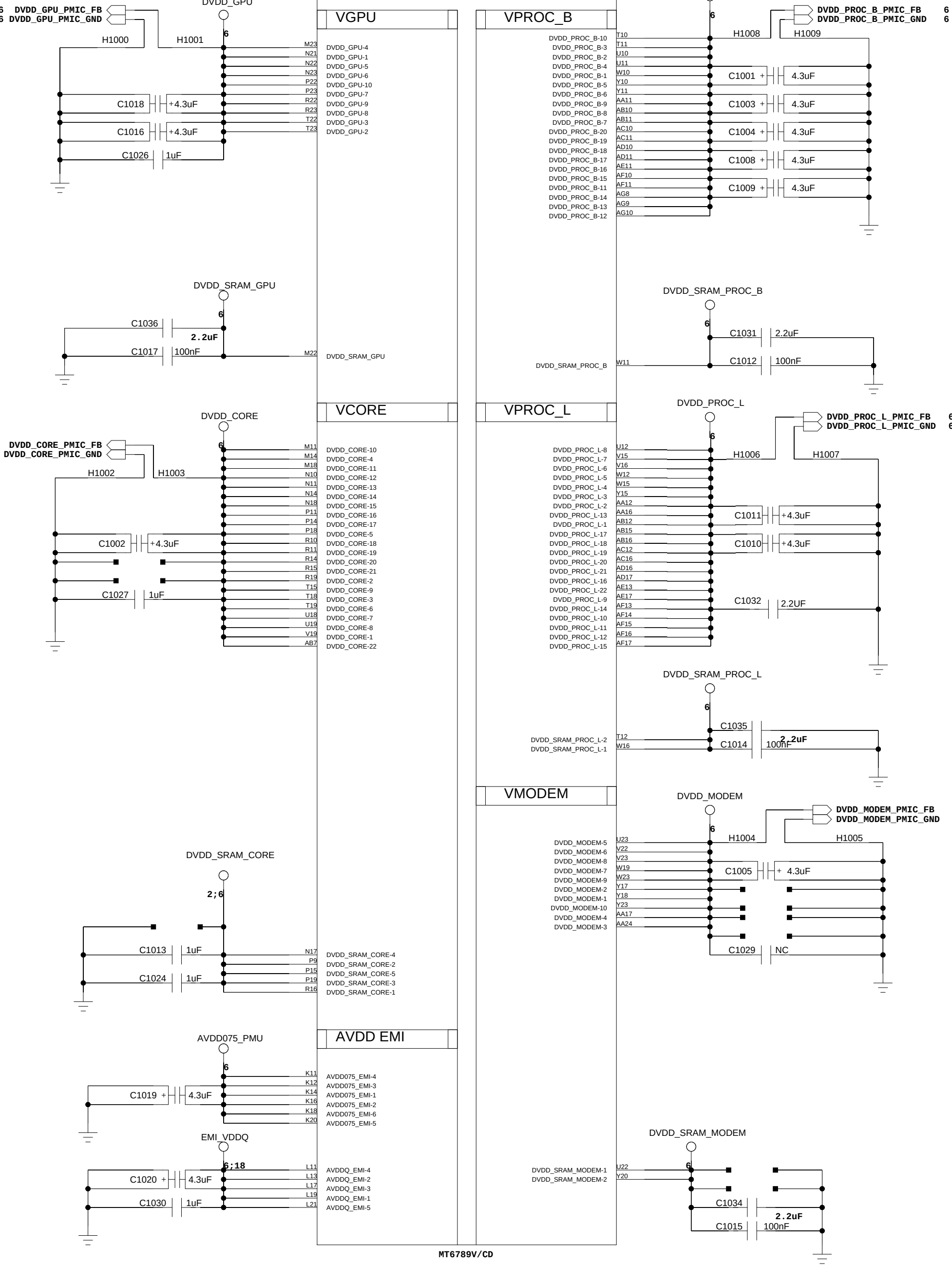
Project : MT6789, LPDDR4X (SBS)

REF_SCH TOP LEVEL

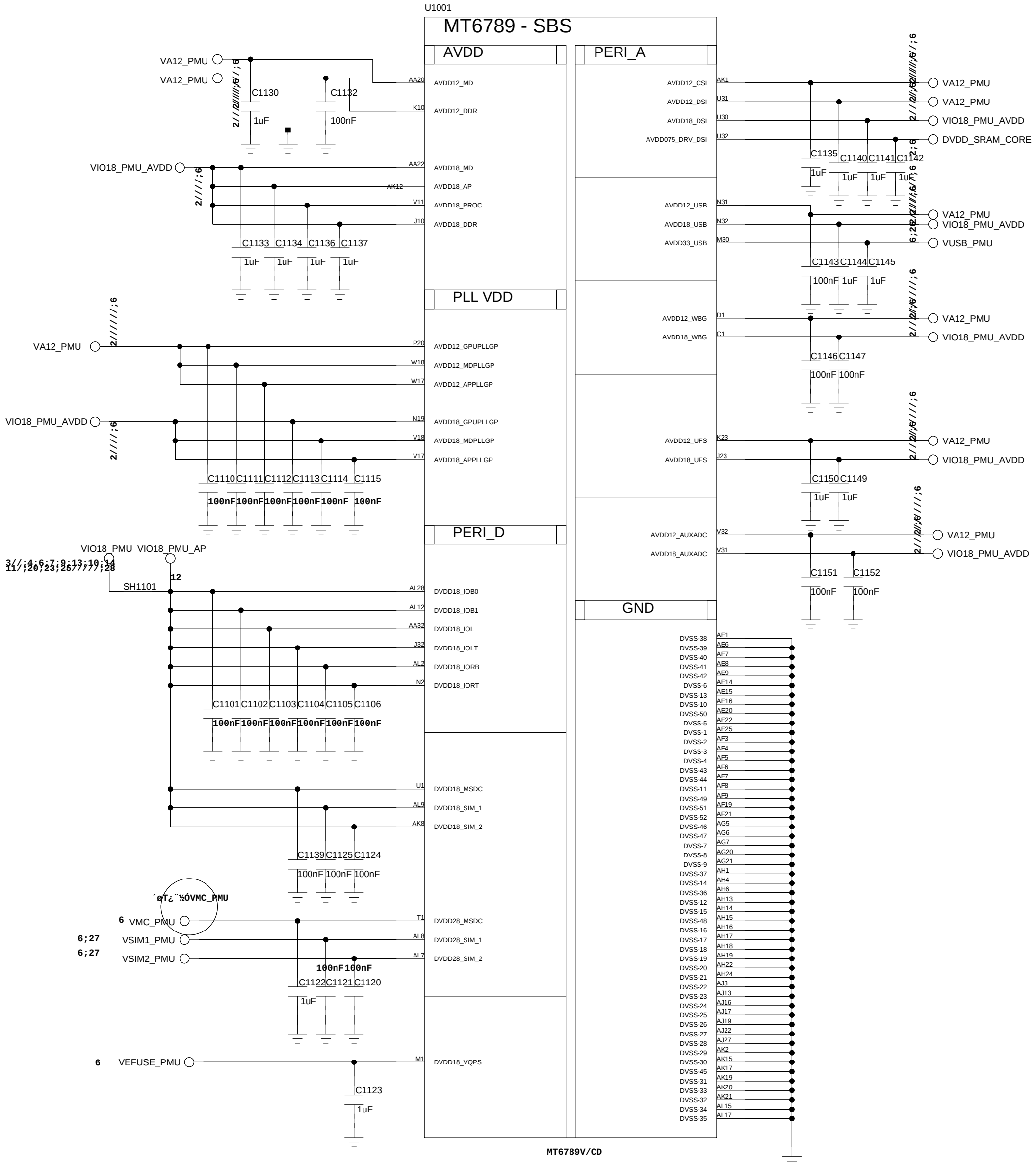


COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 22/05/2023:14:15	
DRAWN		DATED		TITLE: 00_BLOCK_DIAGRAM		VERSION:	SHEET: 1 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

BB_POWER

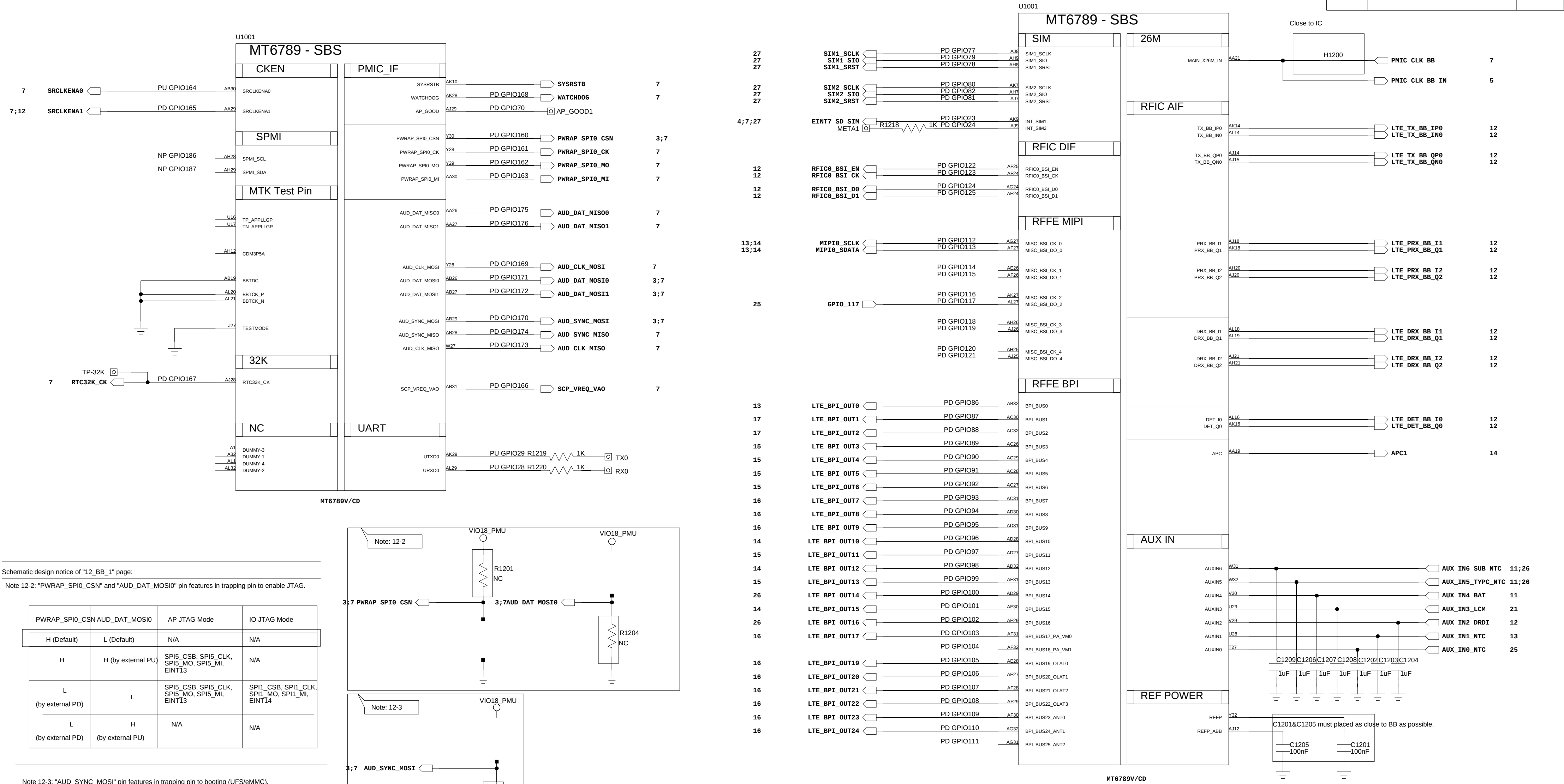


REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

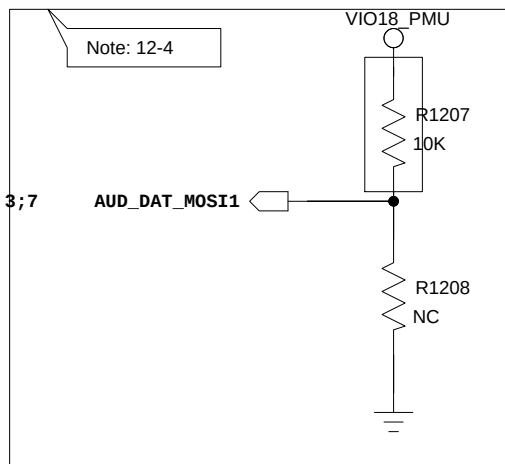
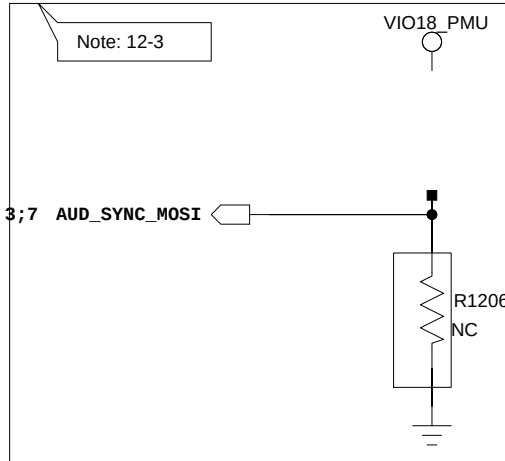
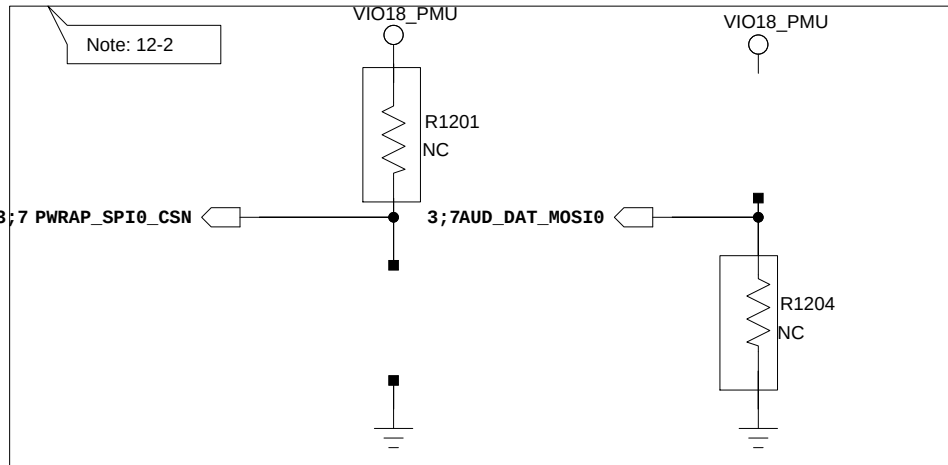


COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 17/06/2023:16:18	
DRAWN		DATED		TITLE: 10_BB_POWER_PDN		VERSION:	
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL	SHEET: 2 OF 28	

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "12_BB_1" page:
Note 12-2: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" pin features in trapping pin to enable JTAG.



Note 12-3: "AUD_SYNC_MOSI" pin features in trapping pin to booting (UFS/eMMC).

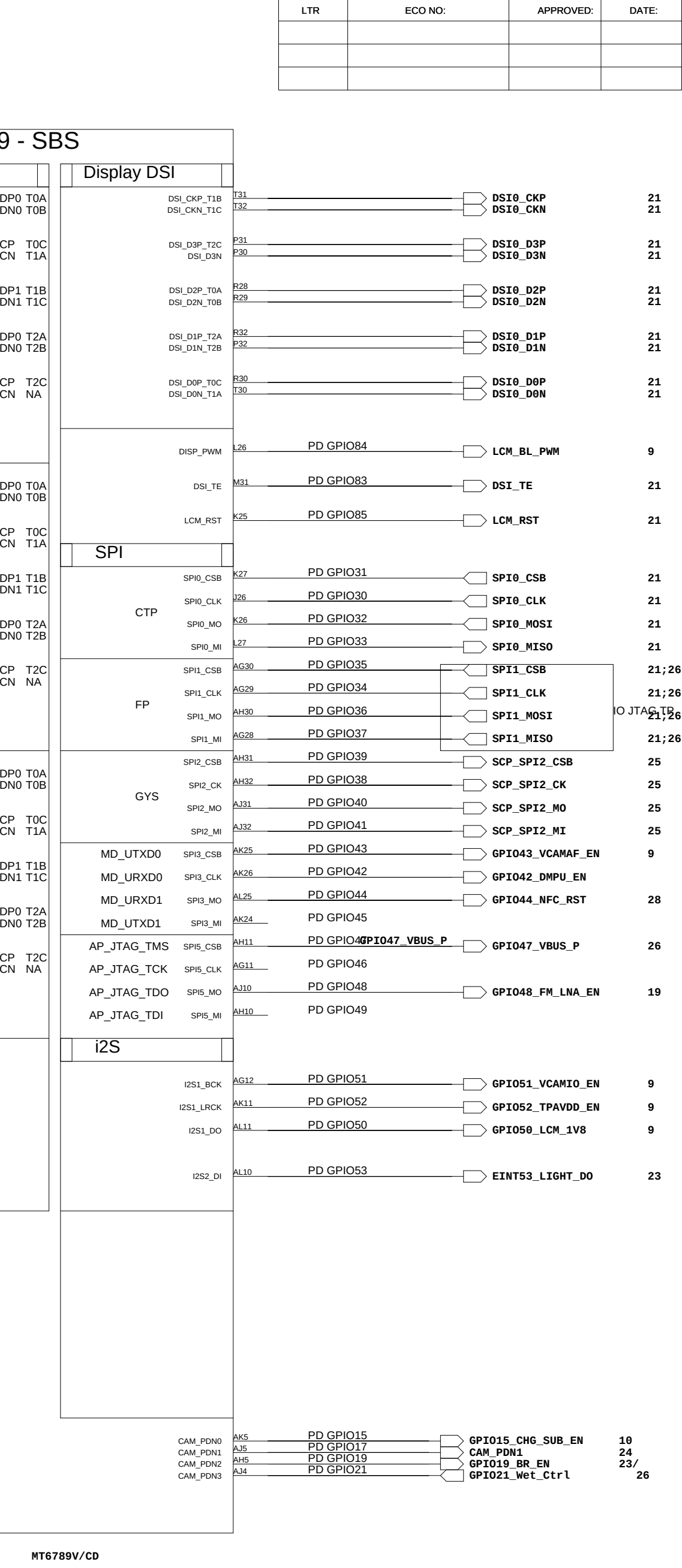
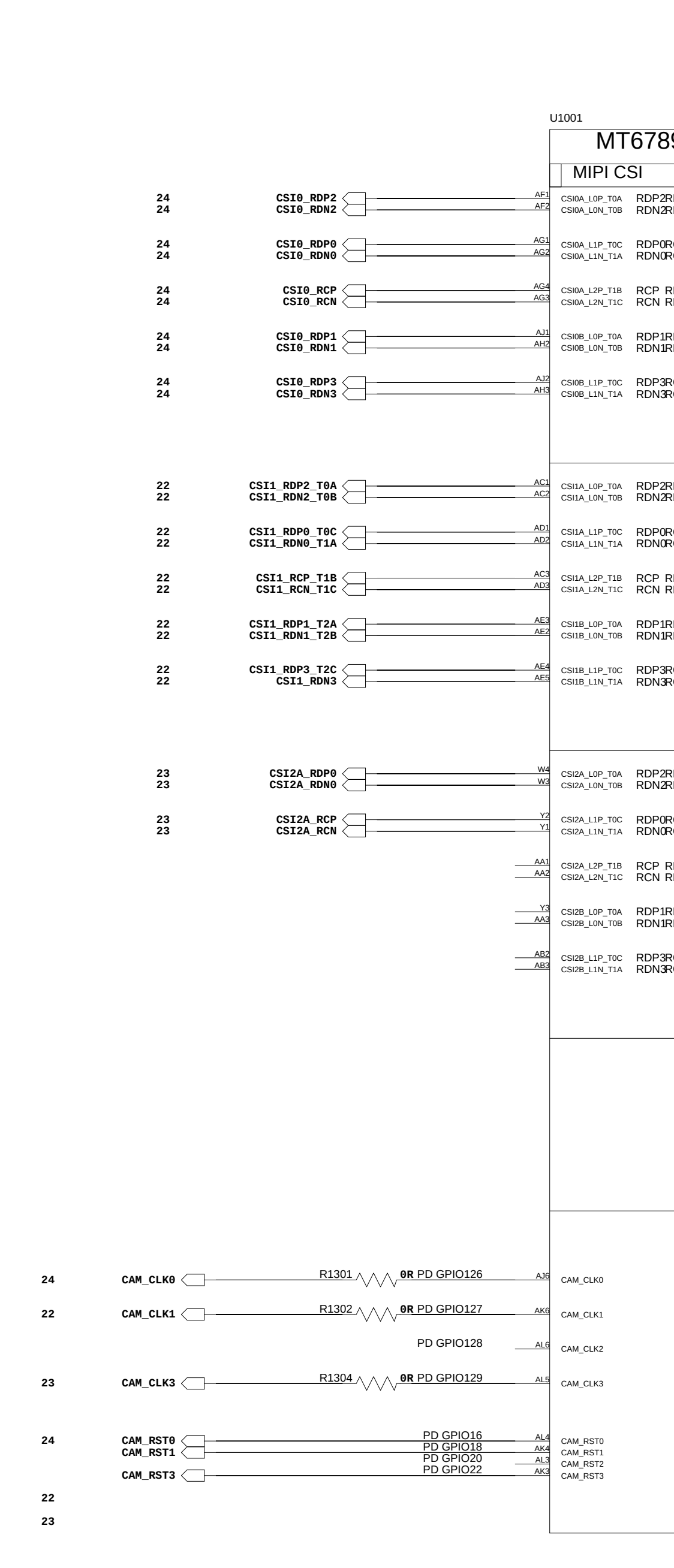
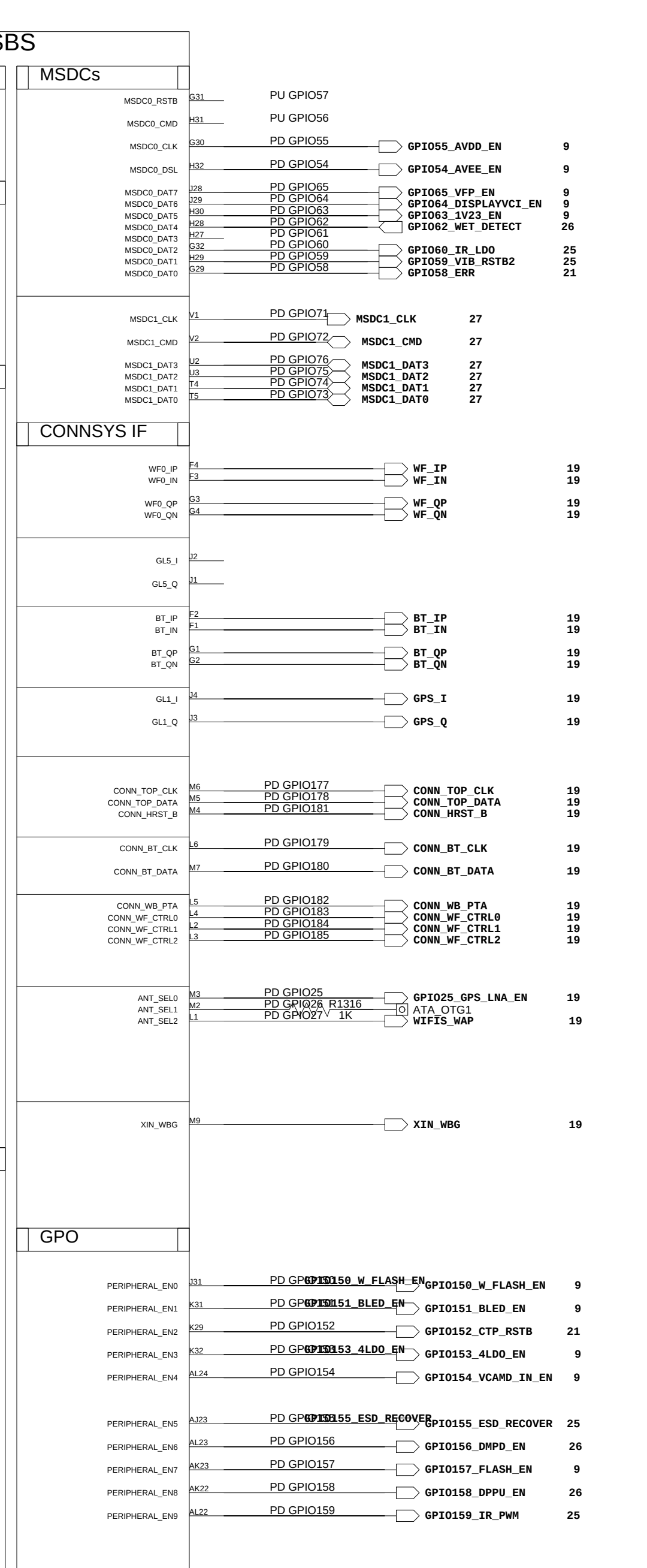
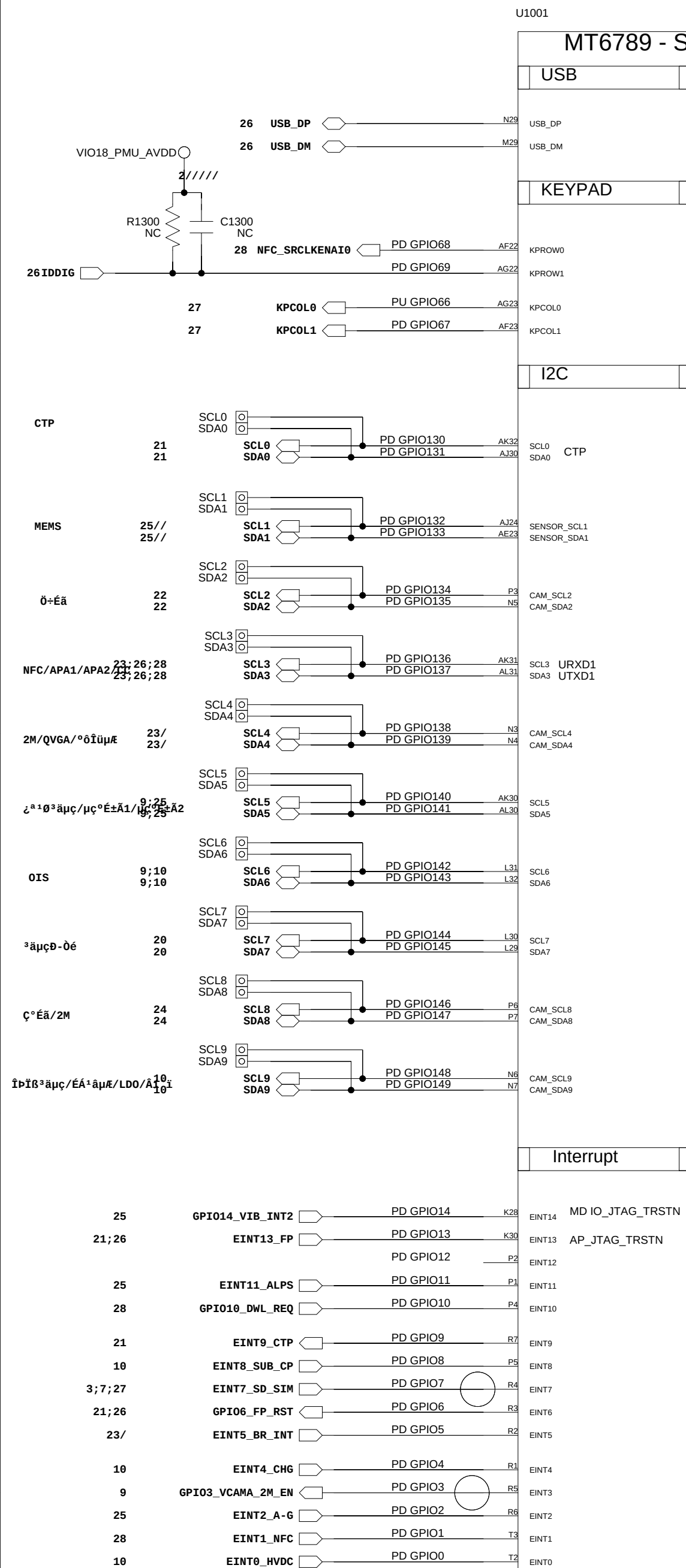
AUD_SYNC_MOSI	Storage Booting Mode
L (Default)	Only UFS boot
H	Only eMMC boot
(by external PU)	

Note 12-4 "AUD_DAT_MOSI1" pin features in trapping pin for SW co-load.

AUD_DAT_MOSI1	Mode
L (Default)	LP4x-MCP
H	LP4x-Discrete
(by external PU)	

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 26/06/2023:18:12	
DRAWN		DATED		TITLE: 12_BB_I_PMIC_RF		VERSION:	SHEET: 3 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

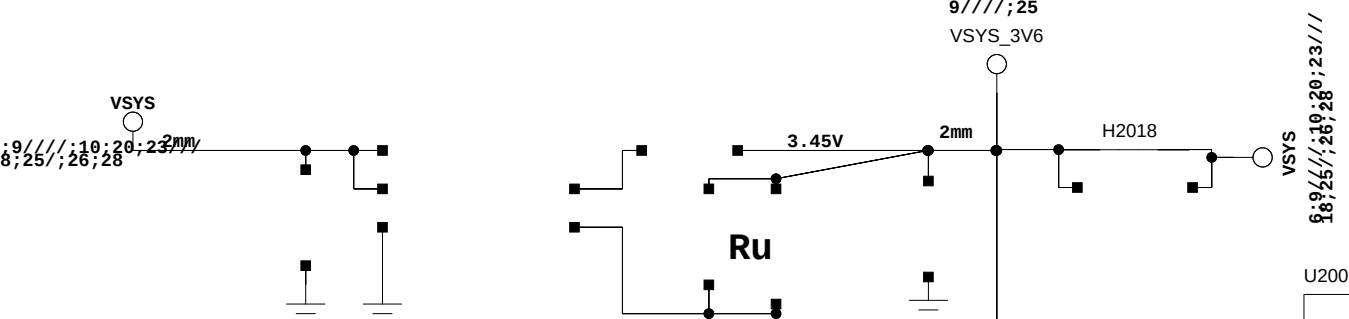
REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 17/06/2023:14:30	
DRAWN		DATED		TITLE: 13_BB_II_CONN_RF		VERSION:	SHEET: 4 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

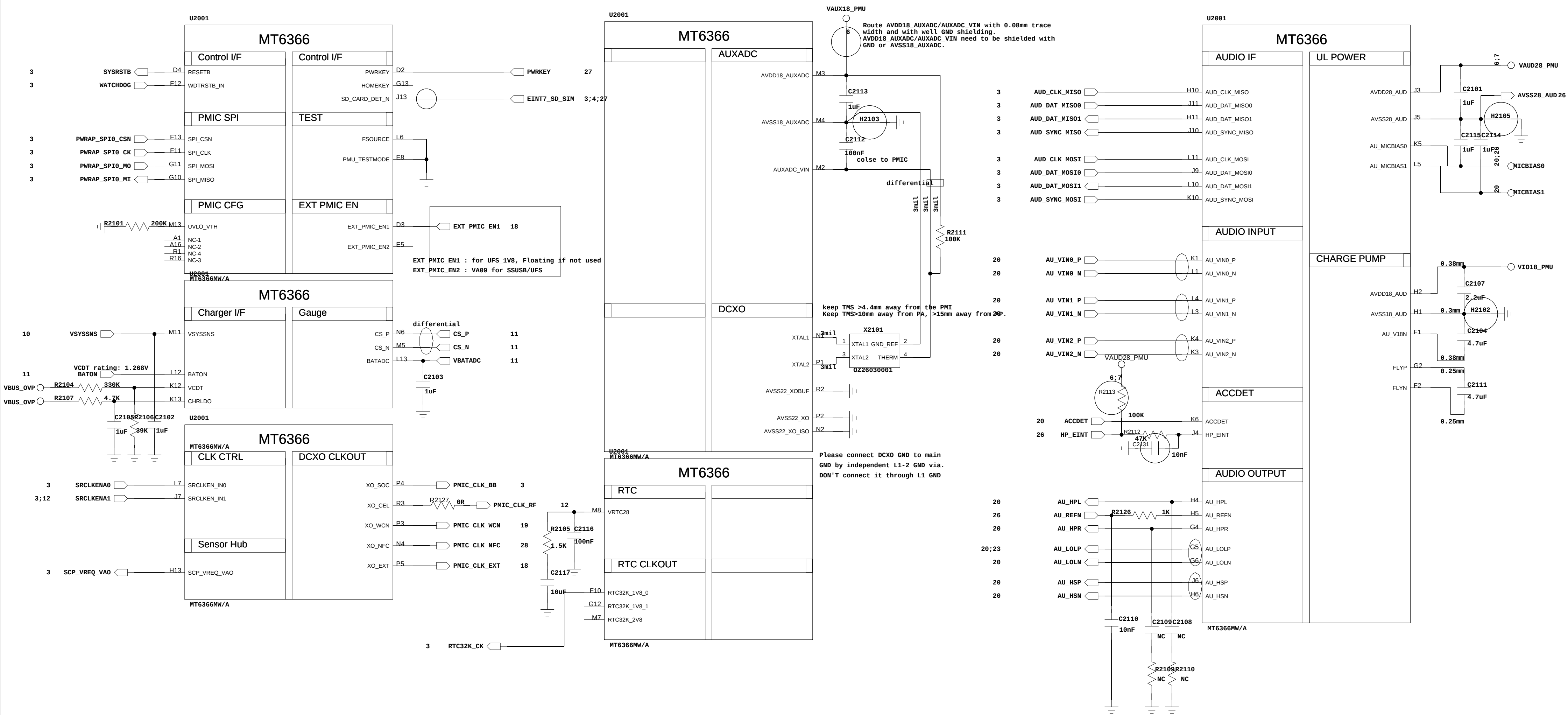
PMU_LDO



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 20/07/2023:21:14	
DRAWN		DATED		TITLE: 20_POWER_MT6366_BUCK_LDO		SHEET: 6 OF 28	
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_MT6366_II

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 16/06/2023:15:07	
DRAWN		DATED		TITLE: 21_POWER_MT6366_GENERAL_CLK		VERSION: SHEET: 7 OF 28	
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

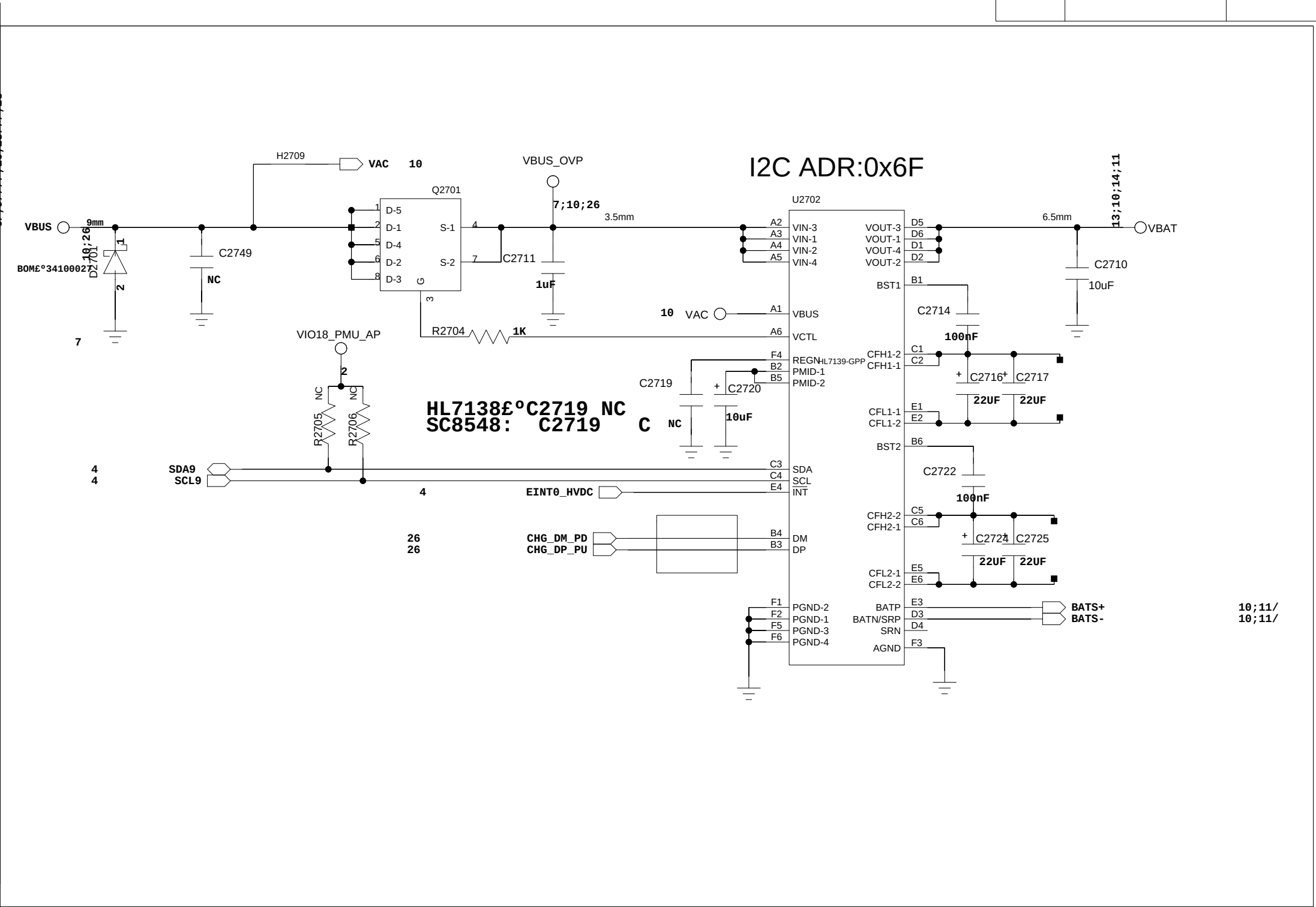
POWER_THIRD-PARTY-I

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

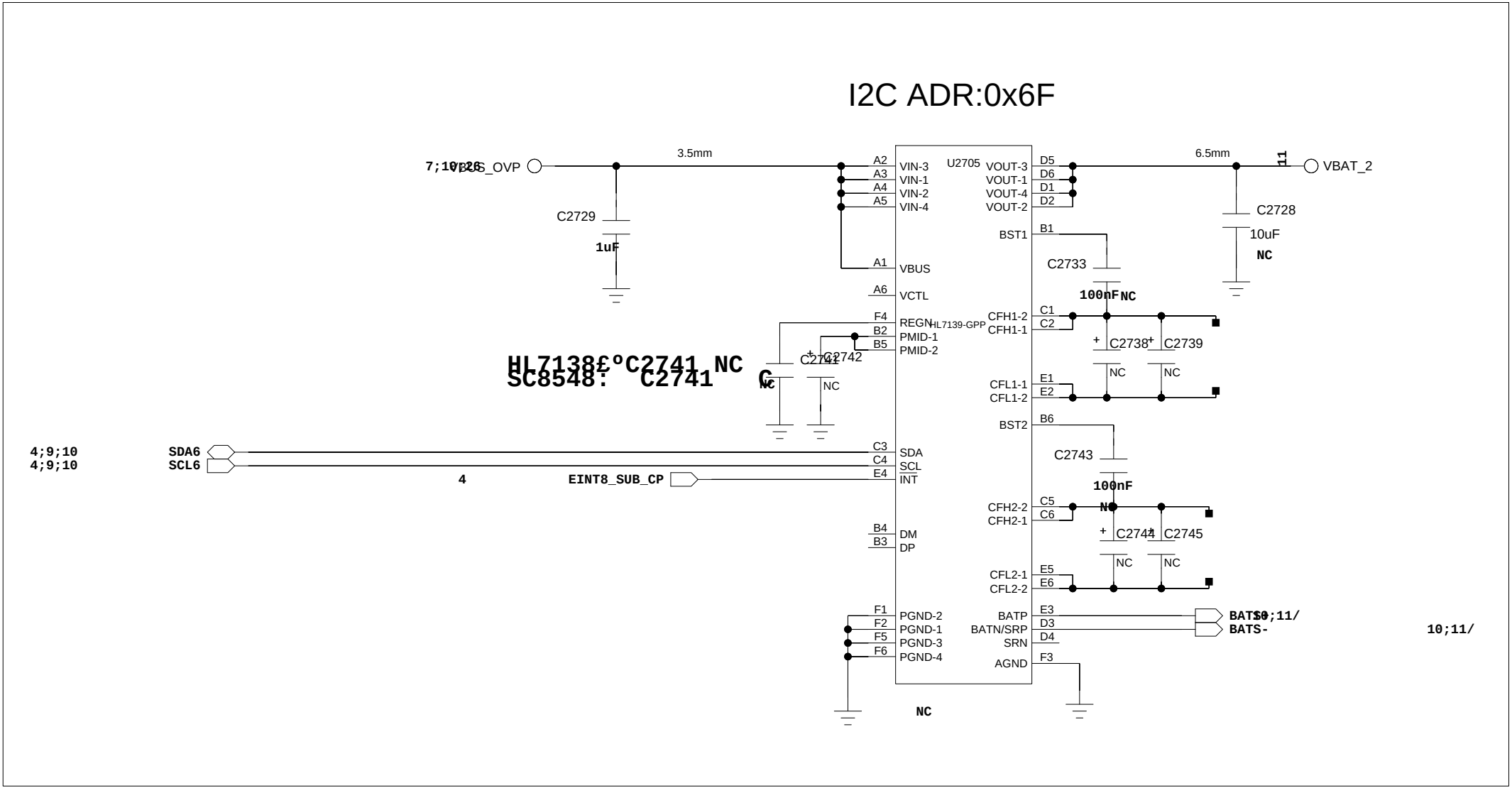
COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 29/05/2023:19:55	
DRAWN		DATED		TITLE: 25_POWER_THIDR-PARTY_I		VERSION:	SHEET: 8 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 21/07/2023:15:22	
DRAWN		DATED		TITLE: 26_POWER_THIRD-PARTY-II		VERSION:	SHEET: 9 OF 28
CHECKED		DATED	< >				

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



- # SUB charger
-
- The schematic diagram illustrates the SUB charger circuit. It features a central IC, U2704 (ETA6937), which manages the charging process. The input side includes a VBUS_CHG pin connected to a network of capacitors (C2734, C2735, C2736) and a 1M resistor. A 10uF capacitor (C2731) is connected to the A1/A2 pins. The output side shows a PL2702 inductor connected to a 10uF capacitor (C2737) and a ground connection. Various control signals are shown, including I2C6_SCL_subcharger, I2C6_SDA_subcharger, and GPIO15_CHG_SUB_EN. The circuit also includes several ground connections (D1-D3, E1-E4) and a VBAT pin.



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 07/06/2023:21:44	
DRAWN		DATED		TITLE: 27_POWER_THIRD-PARTY_III		VERSION:	SHEET: 10 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

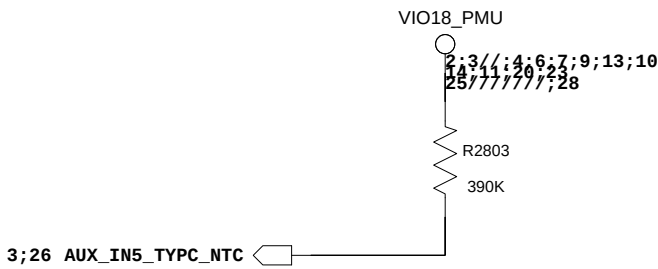
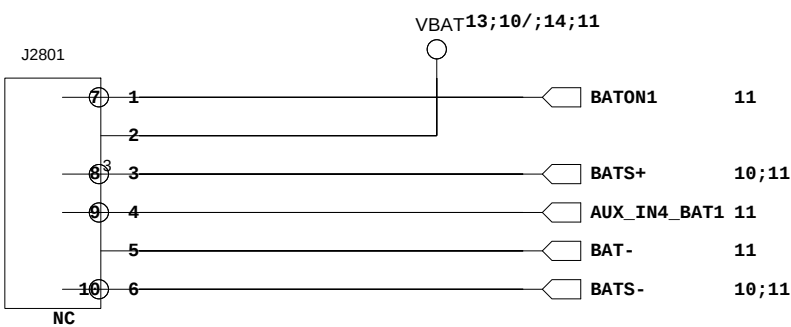
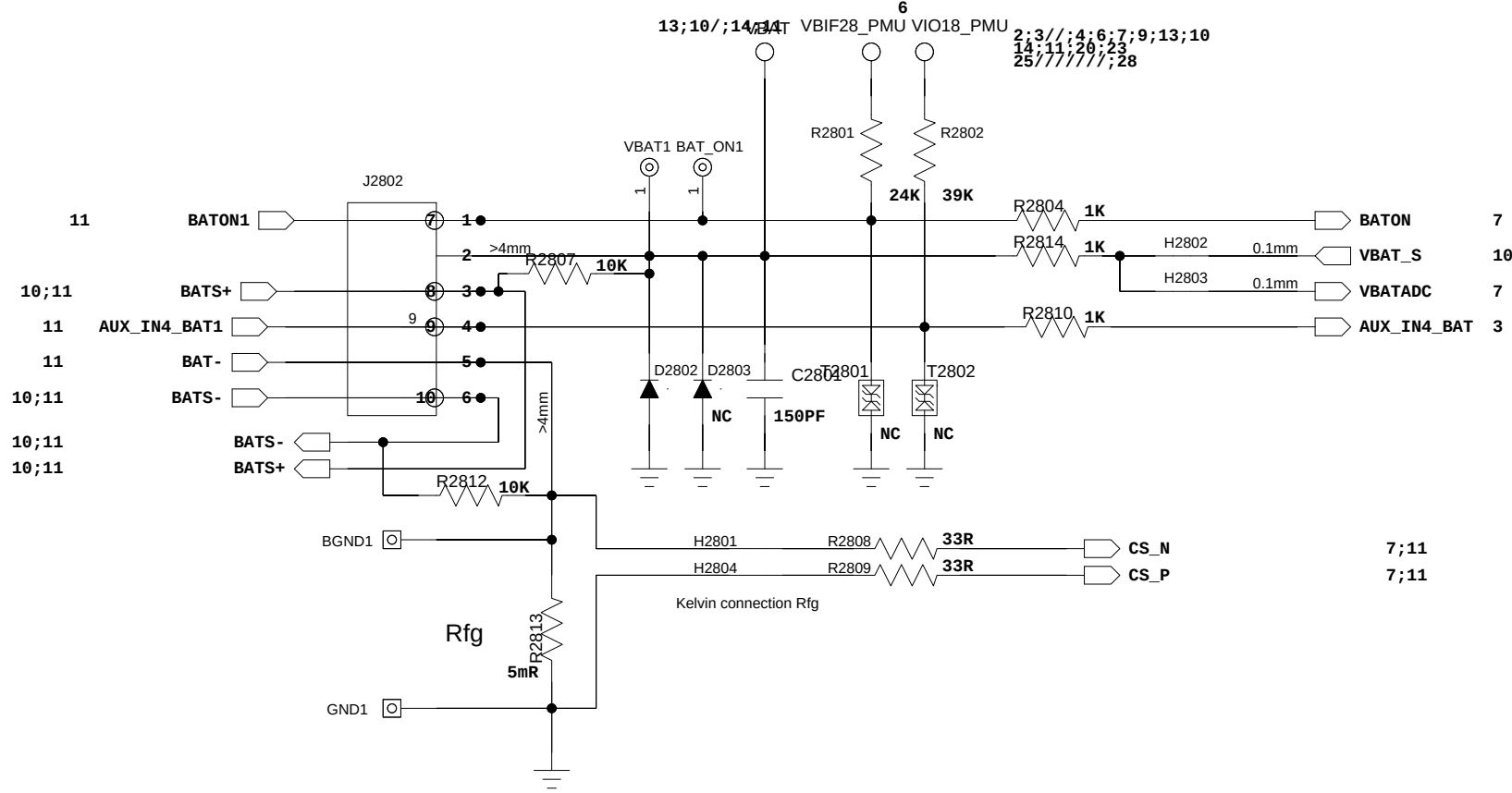
POWER_THIRD-PARTY-IV

WIRELESS CHG

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

Thermistor to sense U2704 temperature

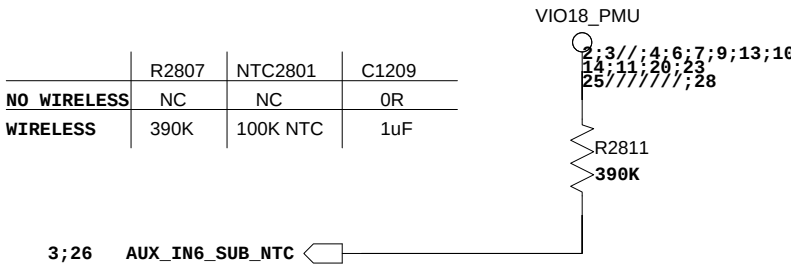
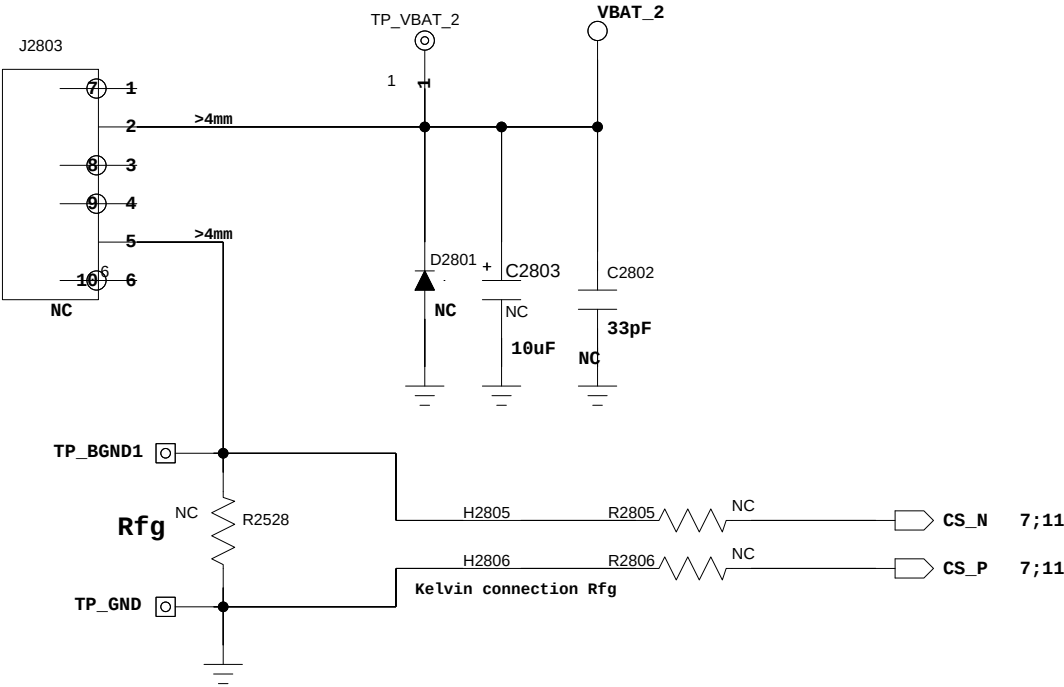
BATTERY CONNECTOR



	R2735	NTC2701
Type-c	NC	NC
Type-c	390K	100K NTC

1. NTC2701 must keep a distance about 2 mm away from U2704 and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

Thermistor to sense U2801 temperature



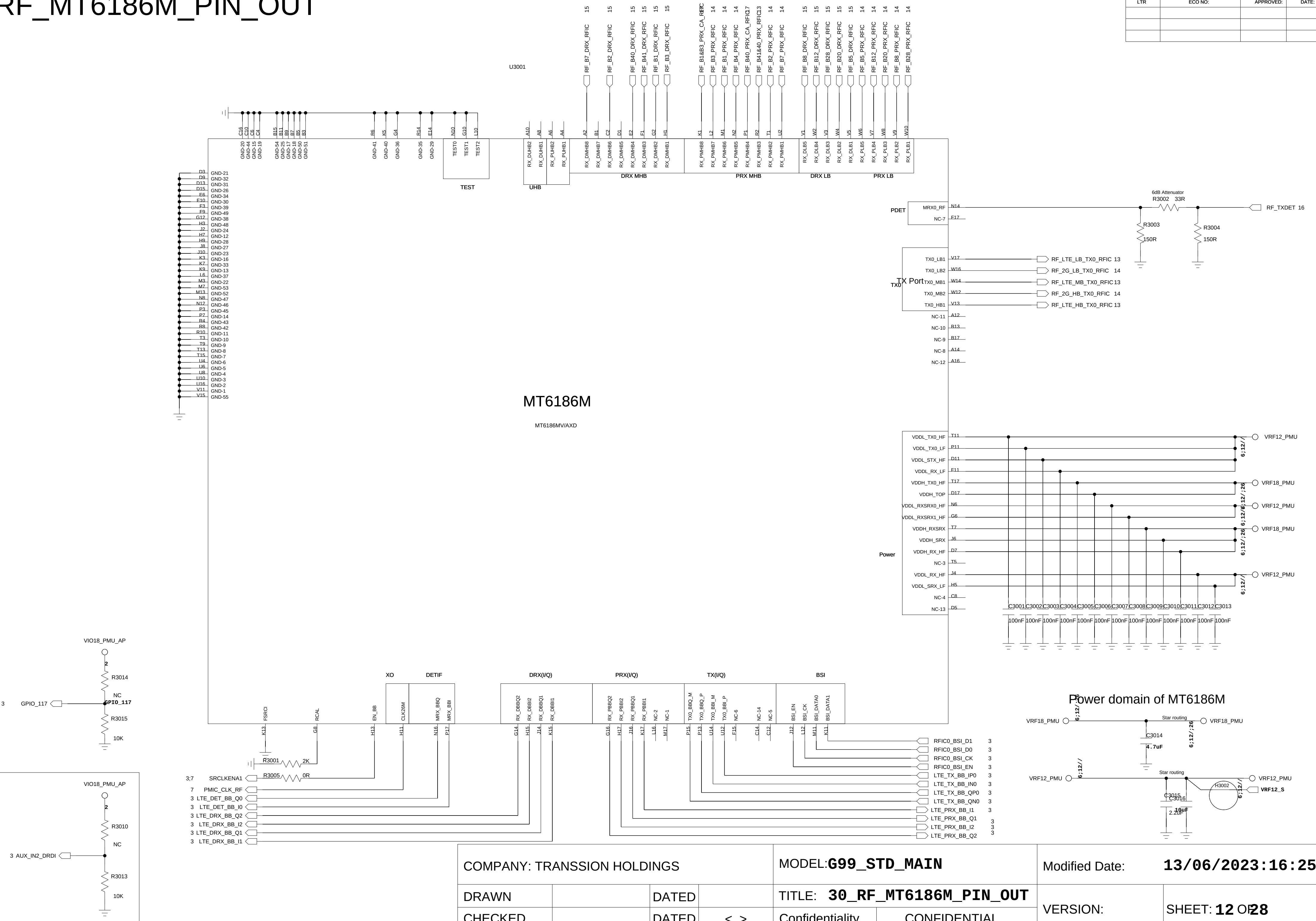
	R2807	NTC2801	C1209
NO WIRELESS	NC	NC	0R
WIRELESS	390K	100K NTC	1uF

1. NTC2801 must keep a distance about 2 mm away from U2801 and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 29/05/2023:19:58	
DRAWN		DATED		TITLE: 28_POWER_THIRD-PARTY-IV		VERSION:	SHEET: 11 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6186M_PIN_OUT

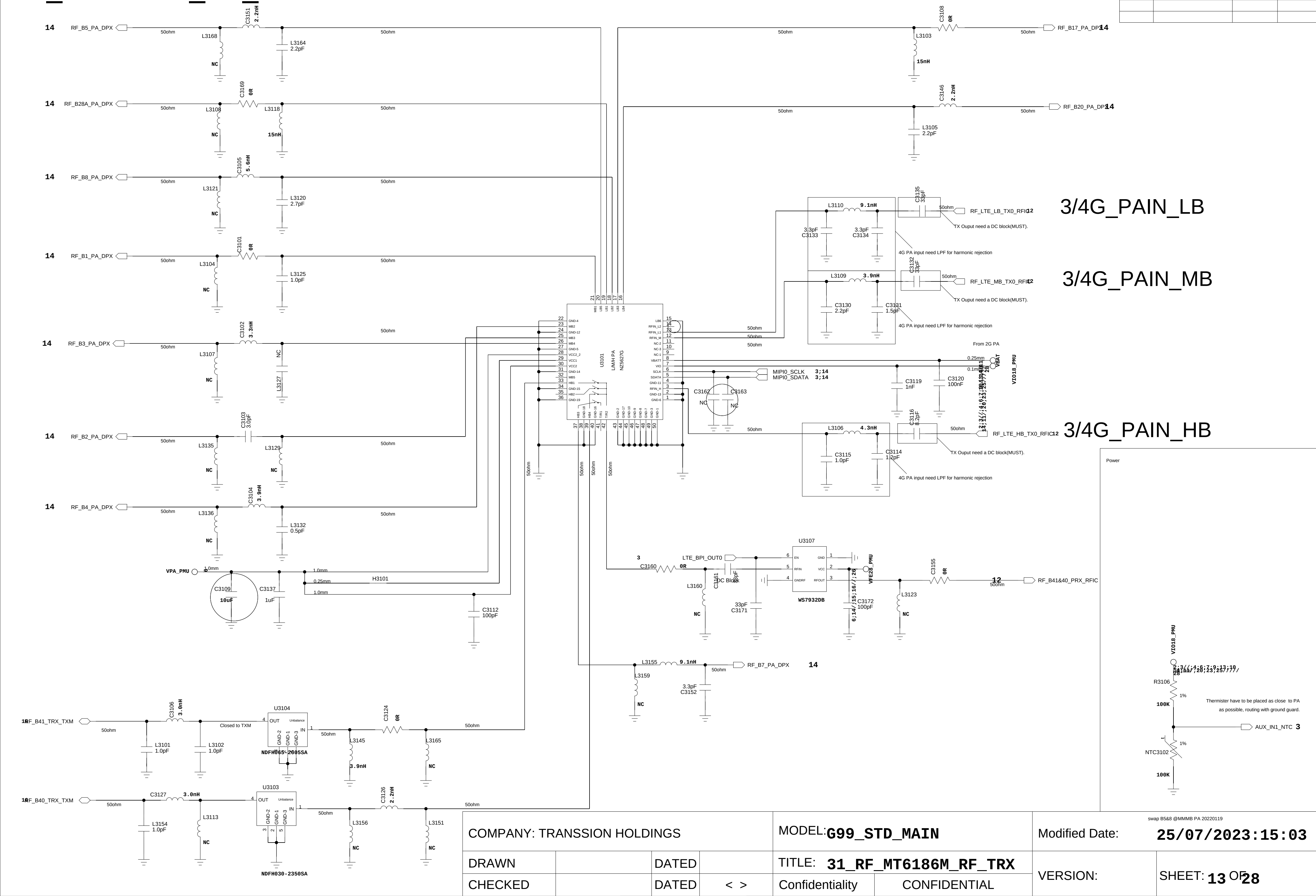
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LTR	ECO NO:	APPROVED:	DATE:



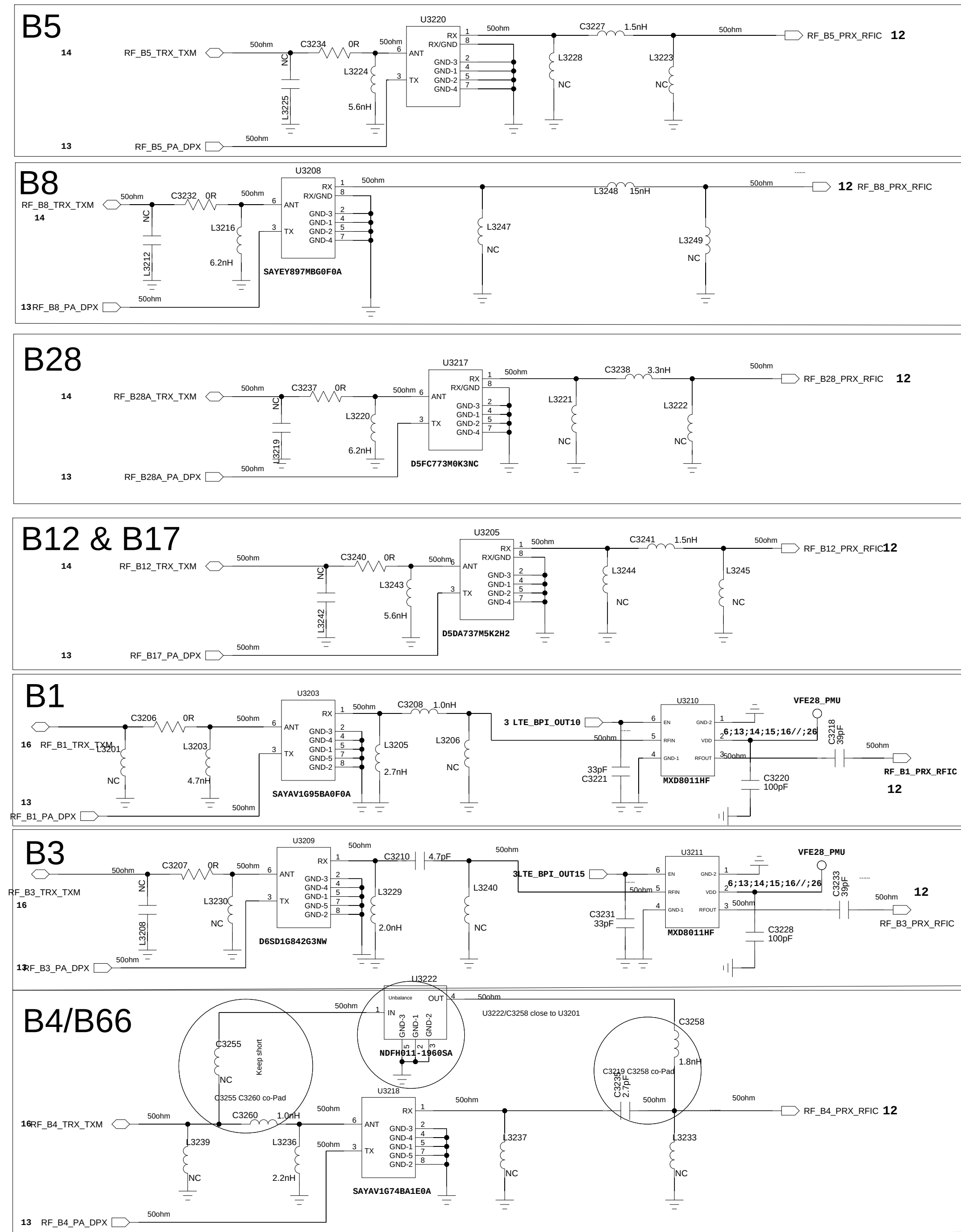
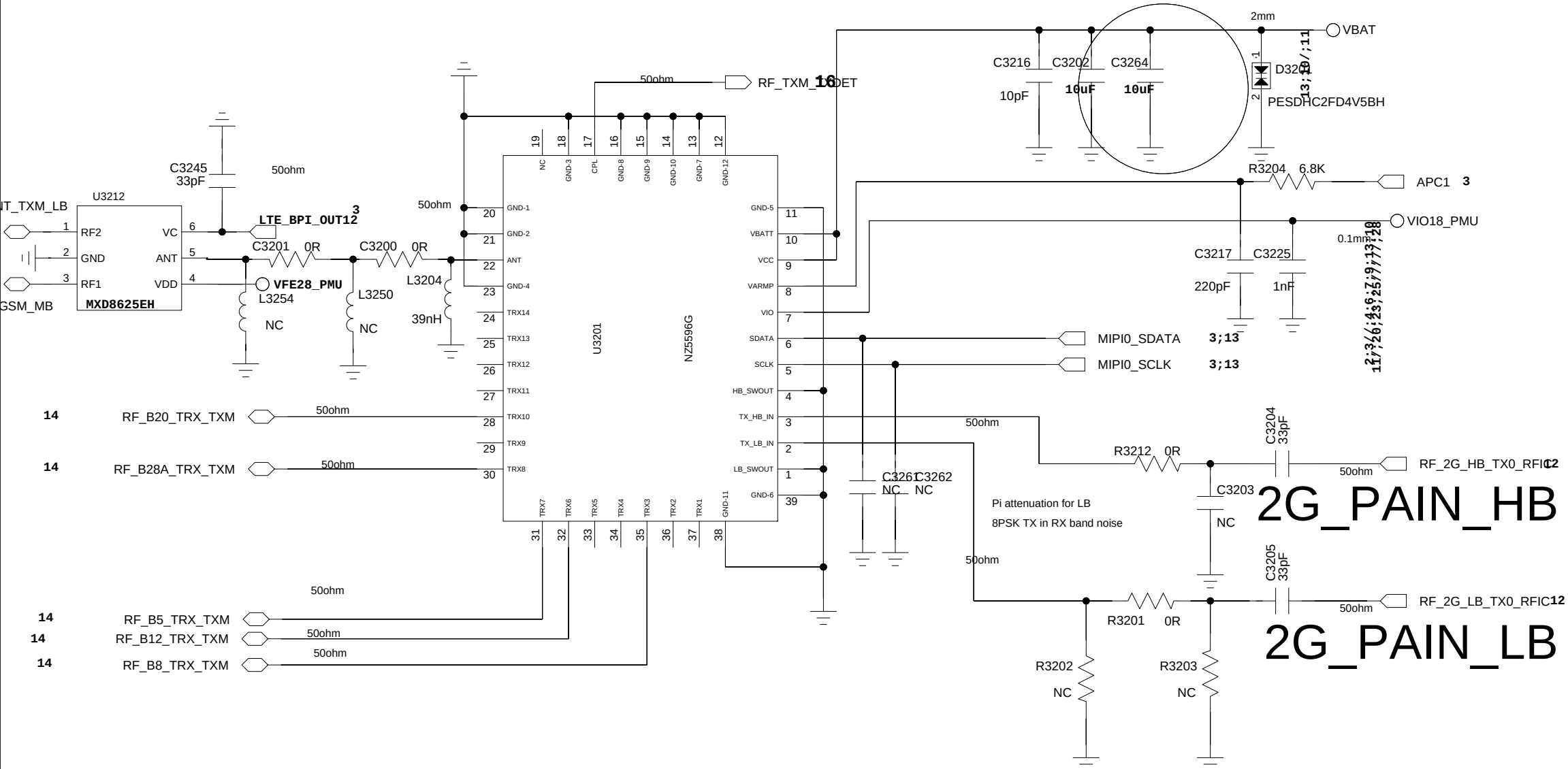
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DRAWN		DATED		TITLE: 30_RF_MT6186M_PIN_OUT		VERSION:	SHEET: 12 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6186M_RF_TX

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

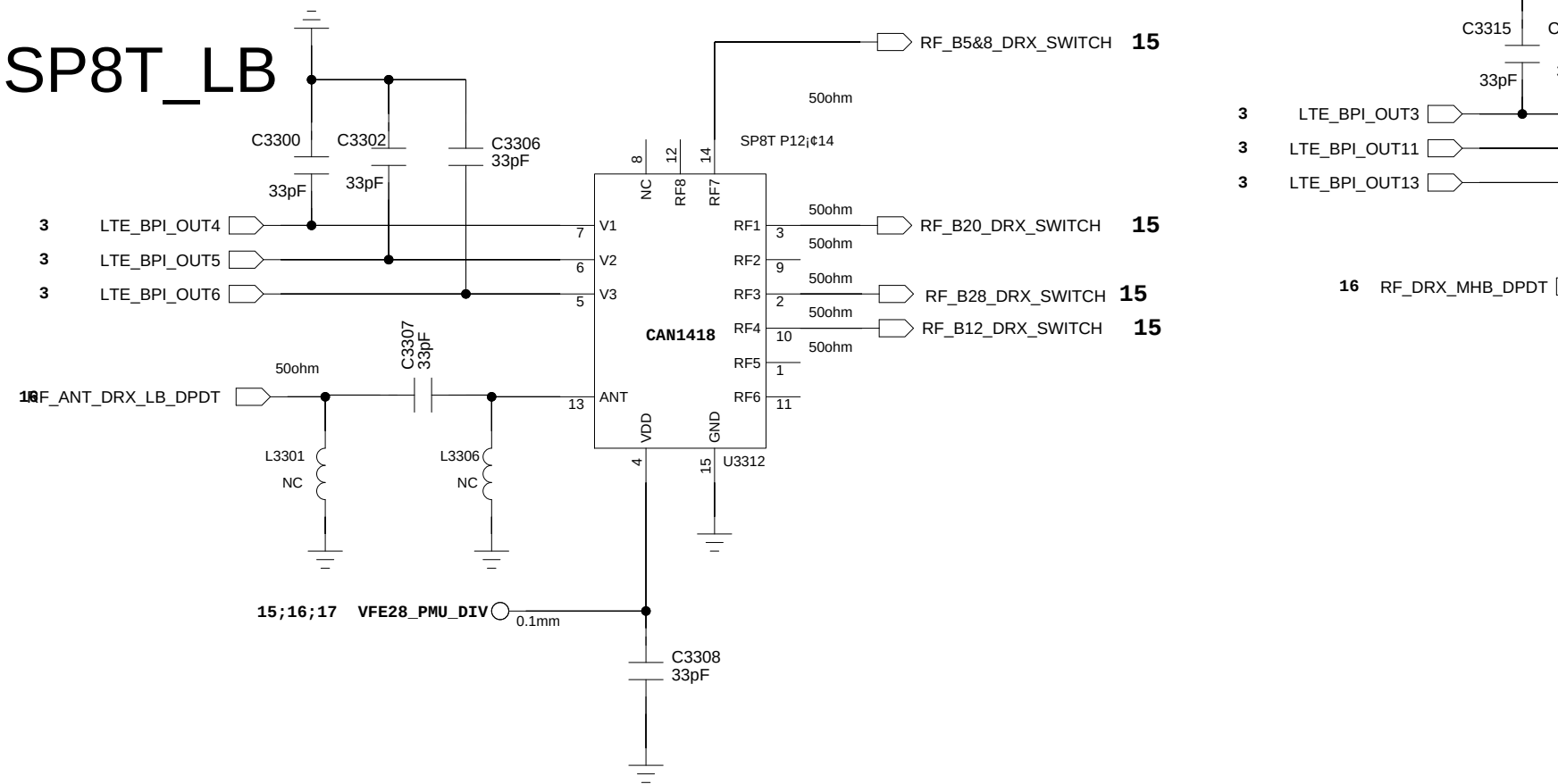


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LTR	ECO NO:	APPROVED:	DATE:

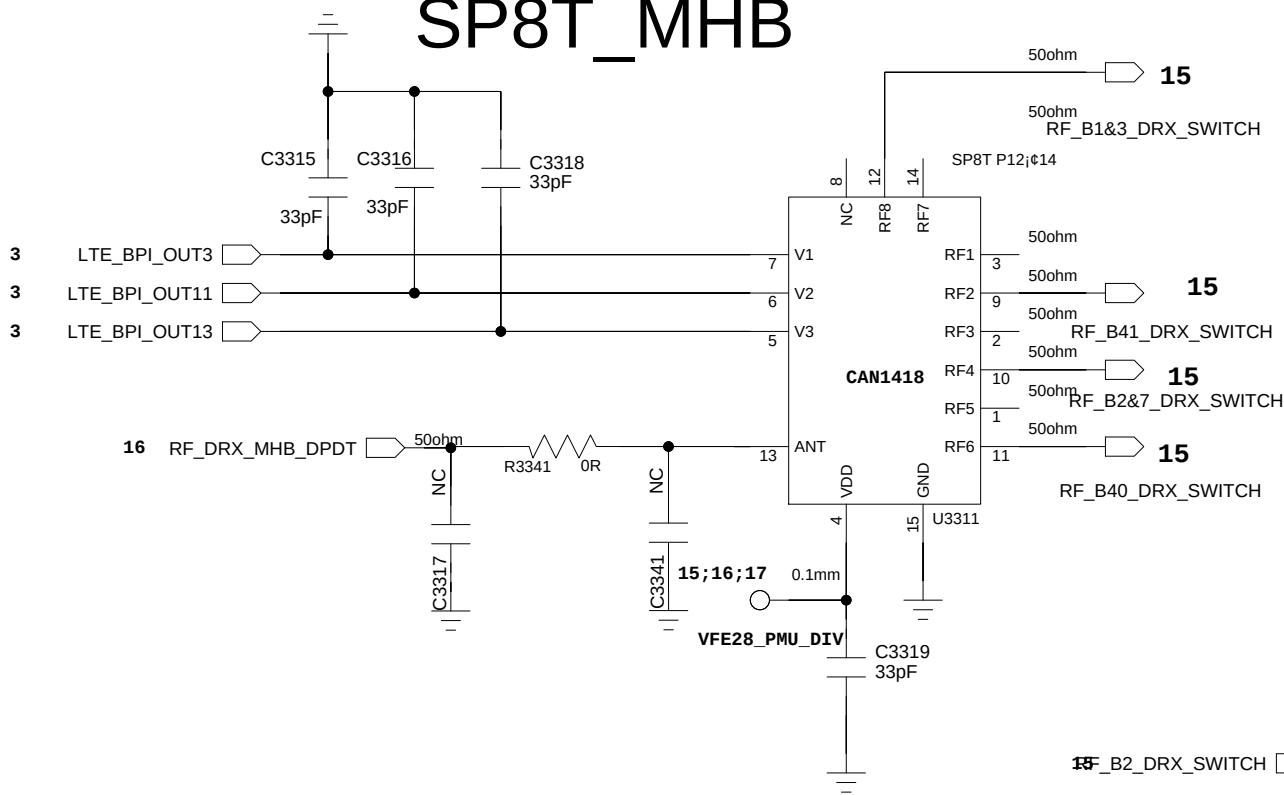


RF_MT6186M_RF_DRX

SP8T_LB

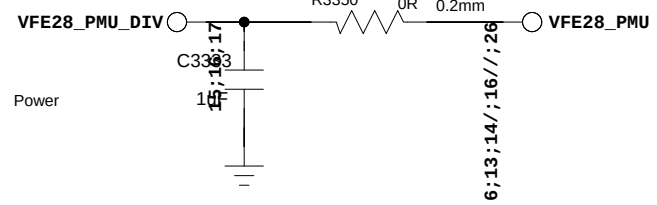


SP8T_MHB

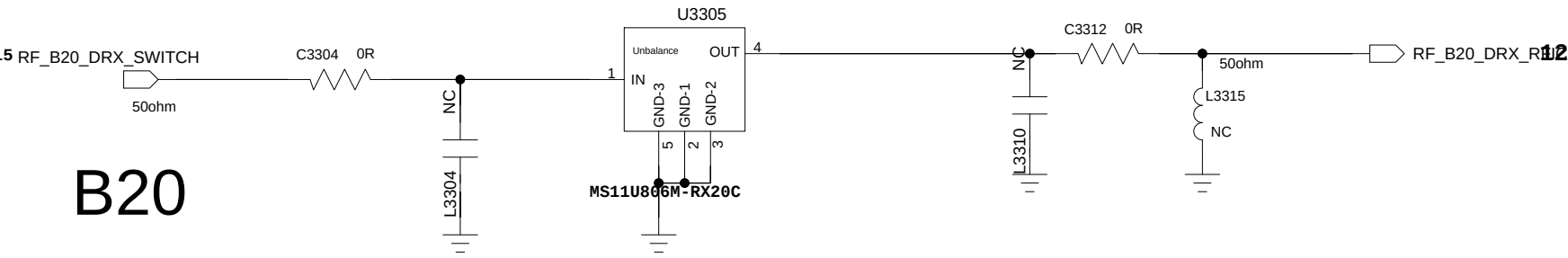


VC1613 control logic					SP8T control logic										
VC1	VC2	RF1	RF2	RF3	SP6T control logic										
V1	V2	V3	RF1	RF2	RF3	RF4	RF5	RF6	RF7	RF8					
H	L	Y	N	N	L	L	L	Y	N	N	N	N	N	N	N
H	H	N	Y	N	L	L	H	N	Y	N	N	N	N	N	N
N	H	N	N	Y	L	H	L	N	N	Y	N	N	N	N	N
					L	H	H	N	N	Y	N	N	N	N	N
					H	L	L	N	N	N	Y	N	N	N	N
					H	L	H	N	N	N	N	Y	N	N	N
					H	H	L	N	N	N	N	N	Y	N	N
					H	H	H	N	N	N	N	N	N	Y	N

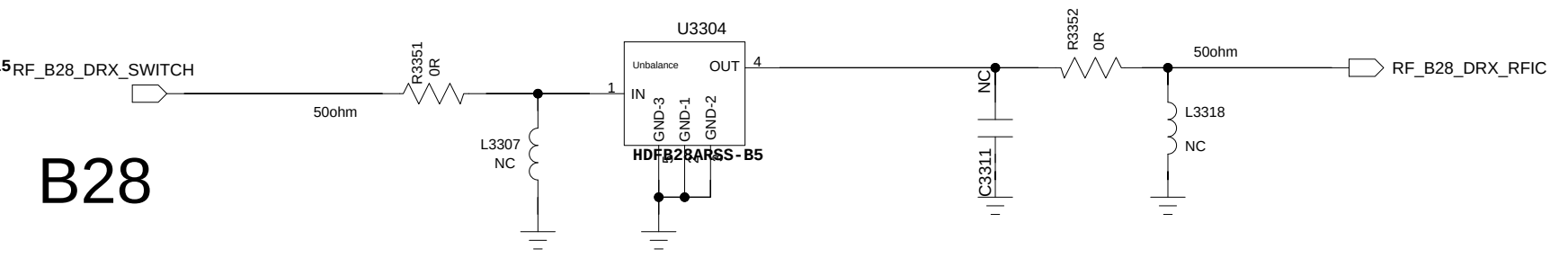
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LTR	ECO NO:	APPROVED:	DATE:



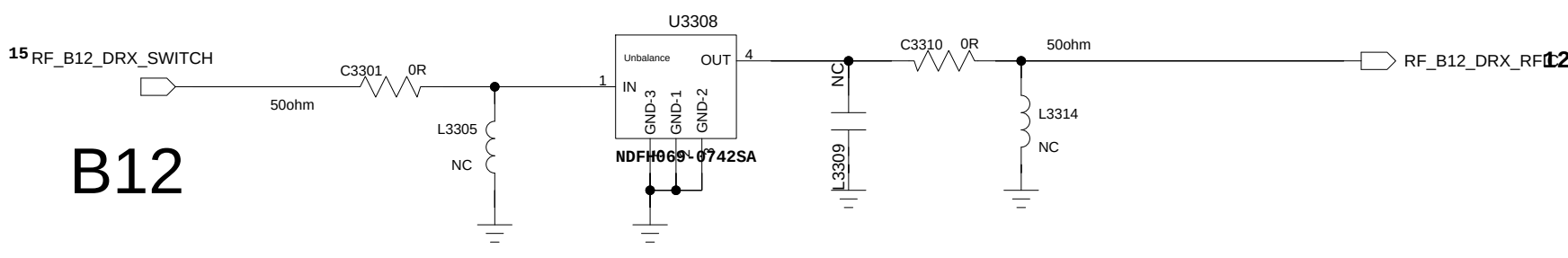
B20



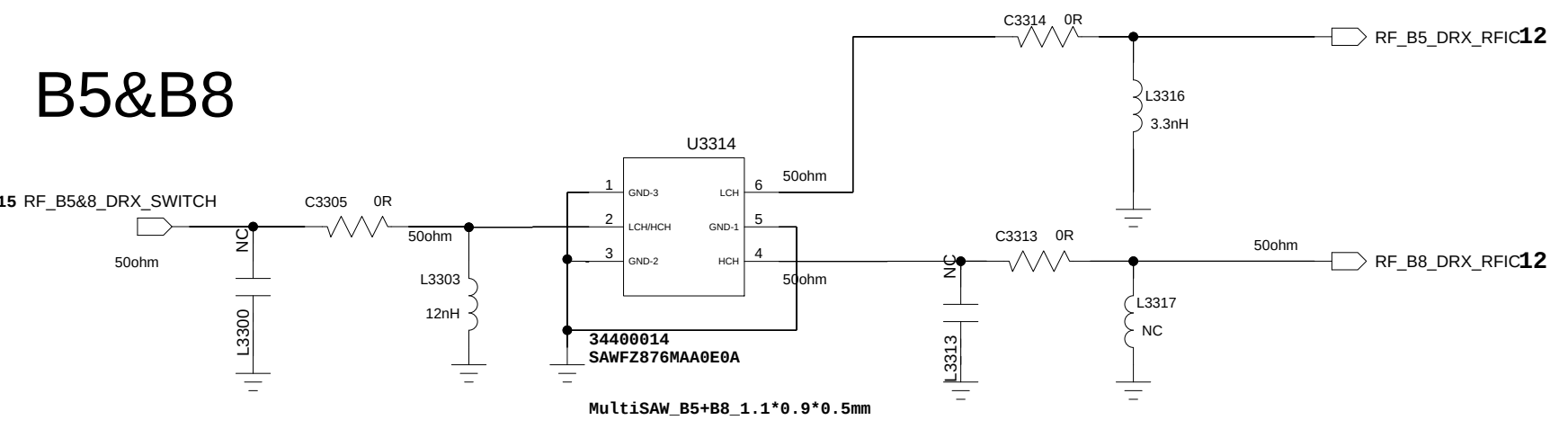
B28



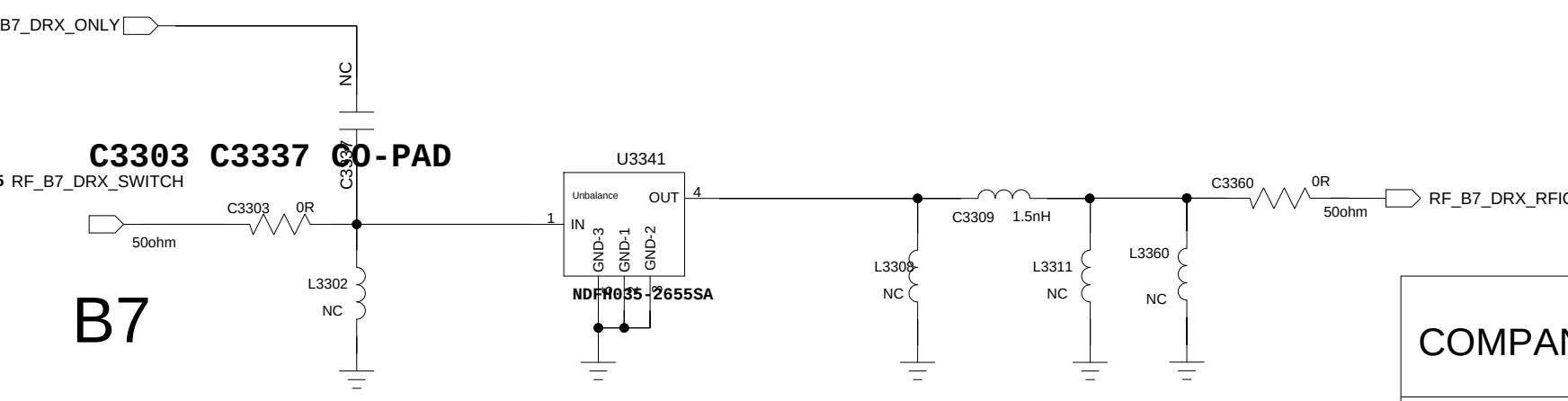
B12



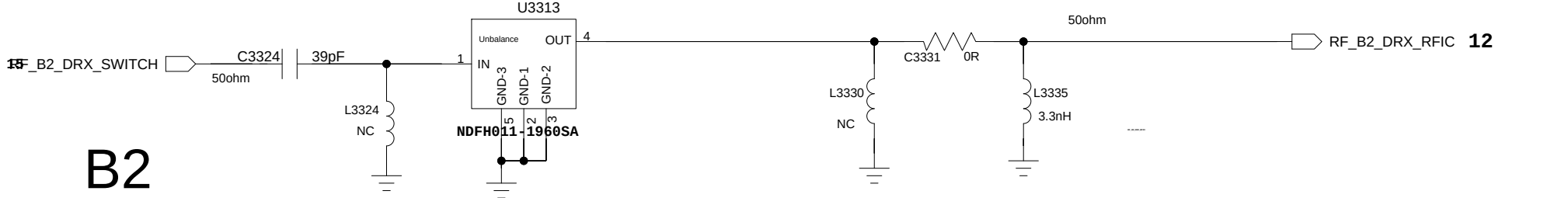
B5&B8



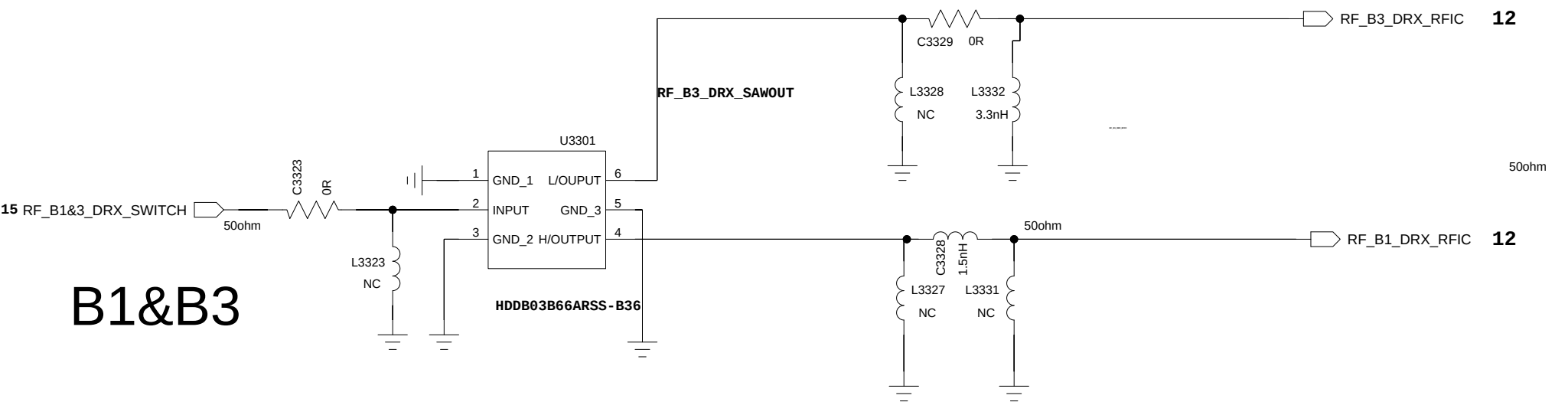
B7



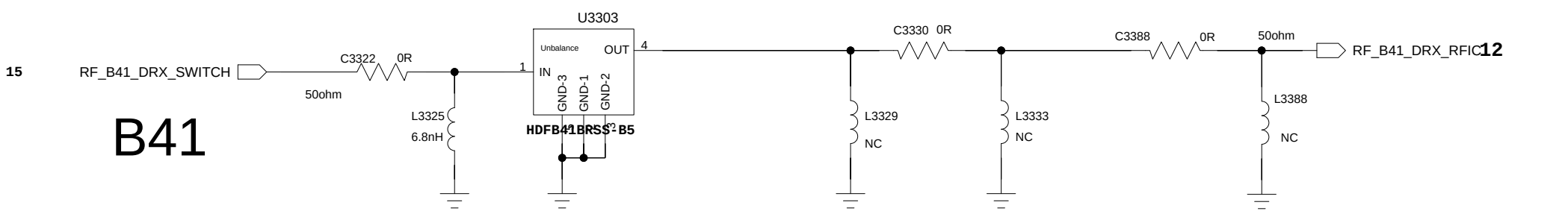
B2



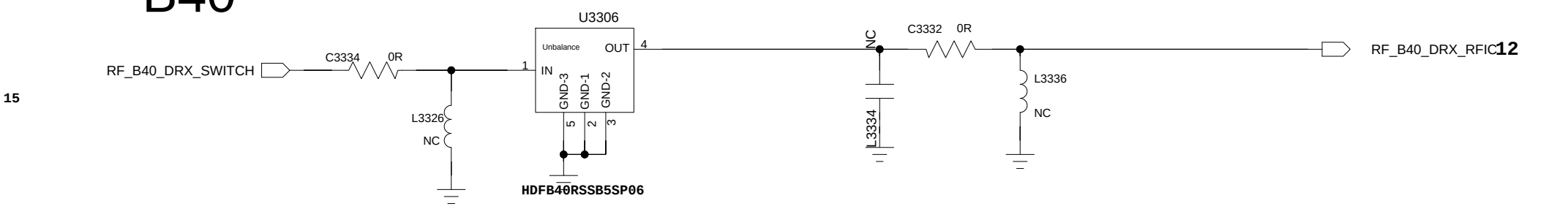
B1&B3



B41



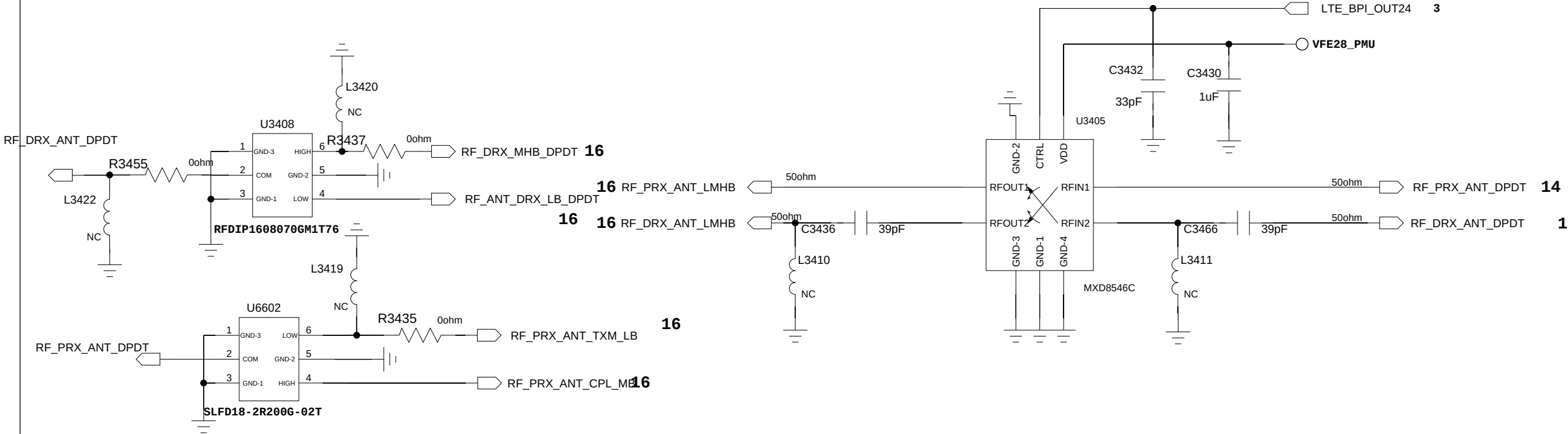
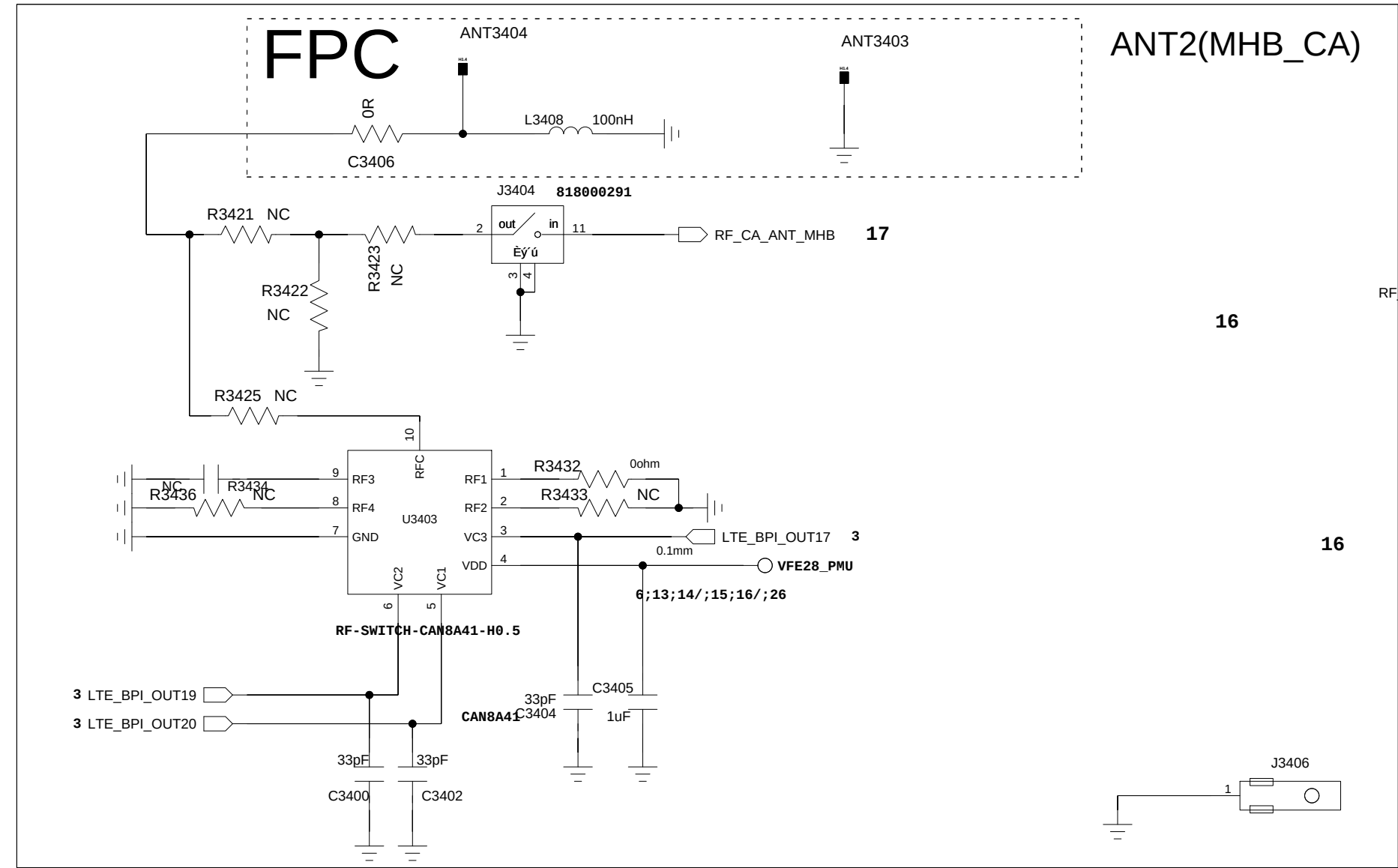
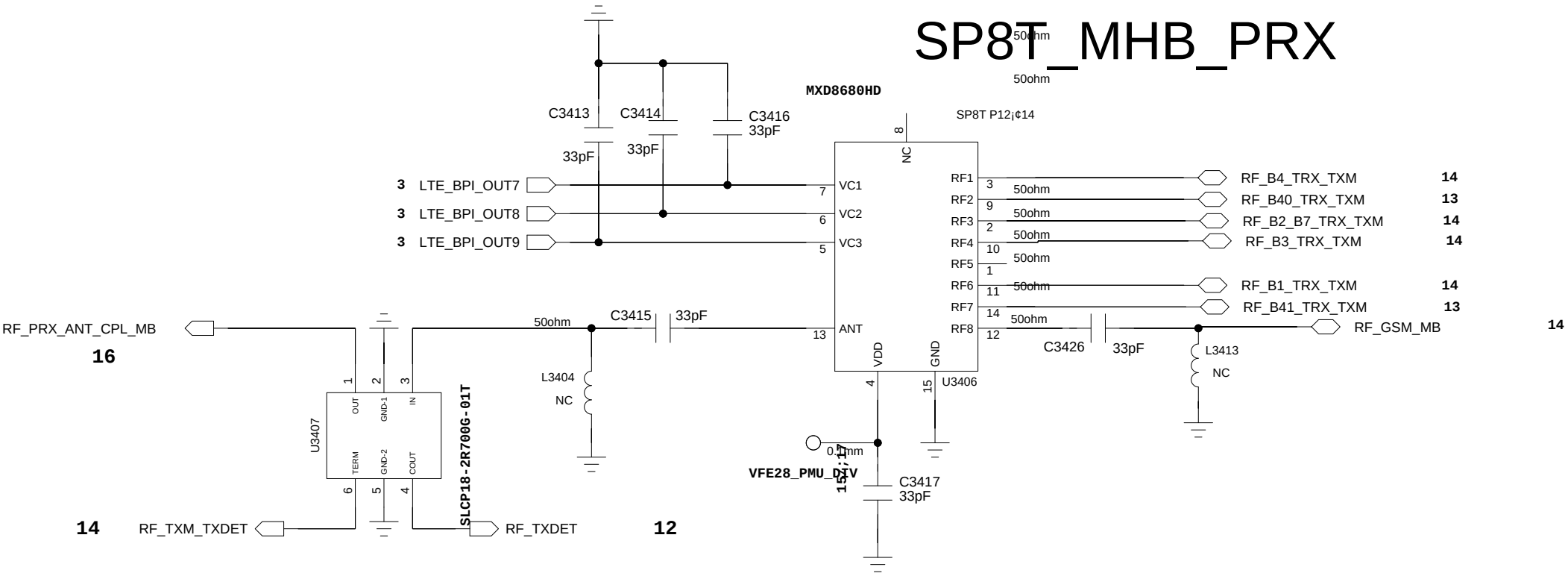
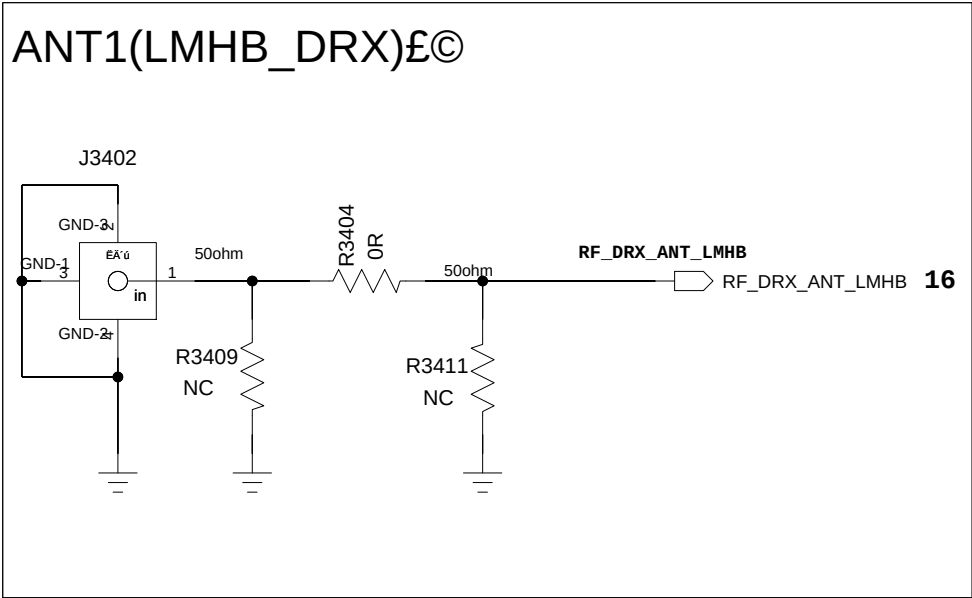
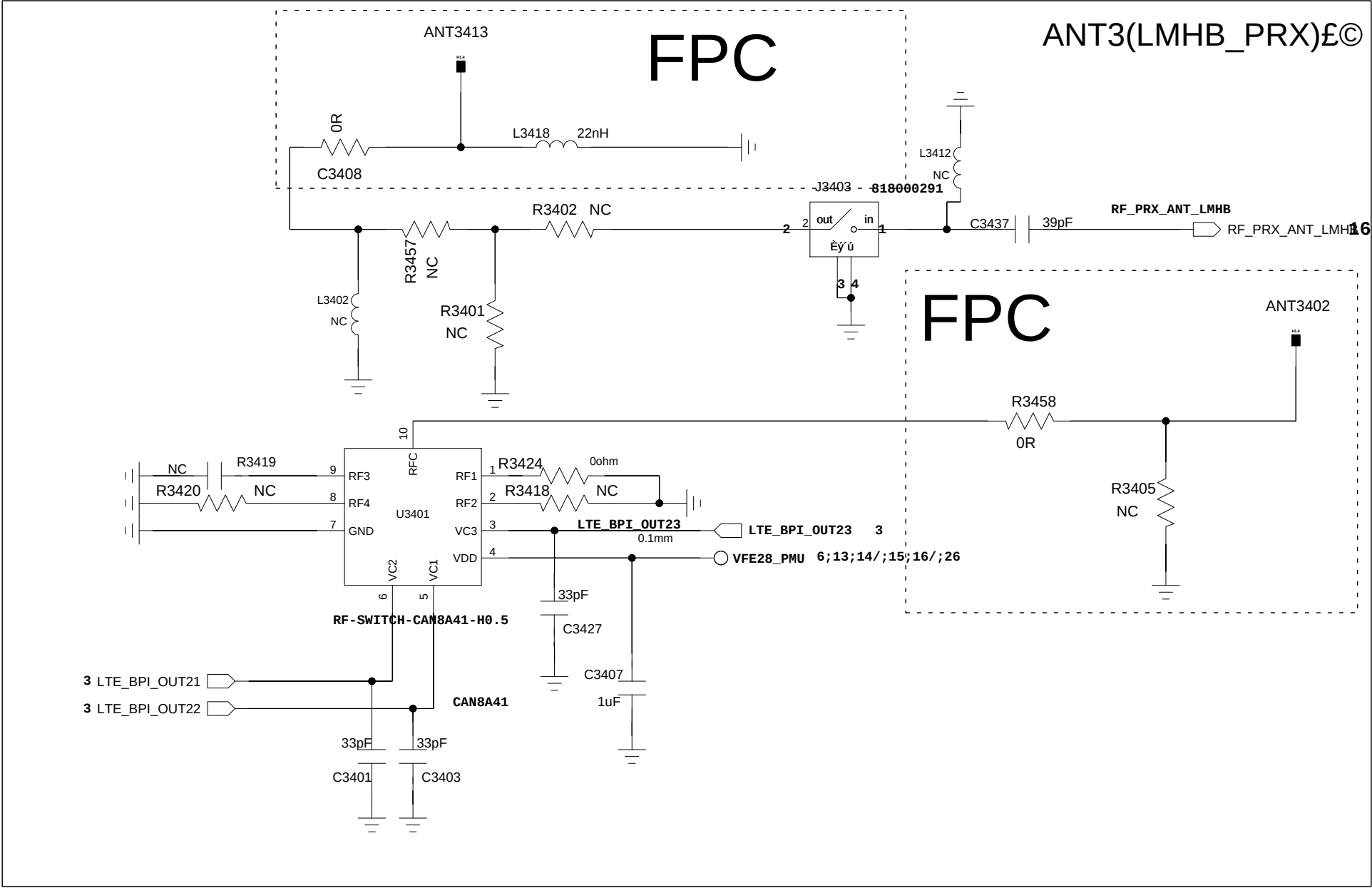
B40



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 20/07/2023:20:58	
DRAWN		DATED		TITLE: 33_RF_MT6186M_RF_DRX		VERSION:	SHEET: 15 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6186M_RF_ANT

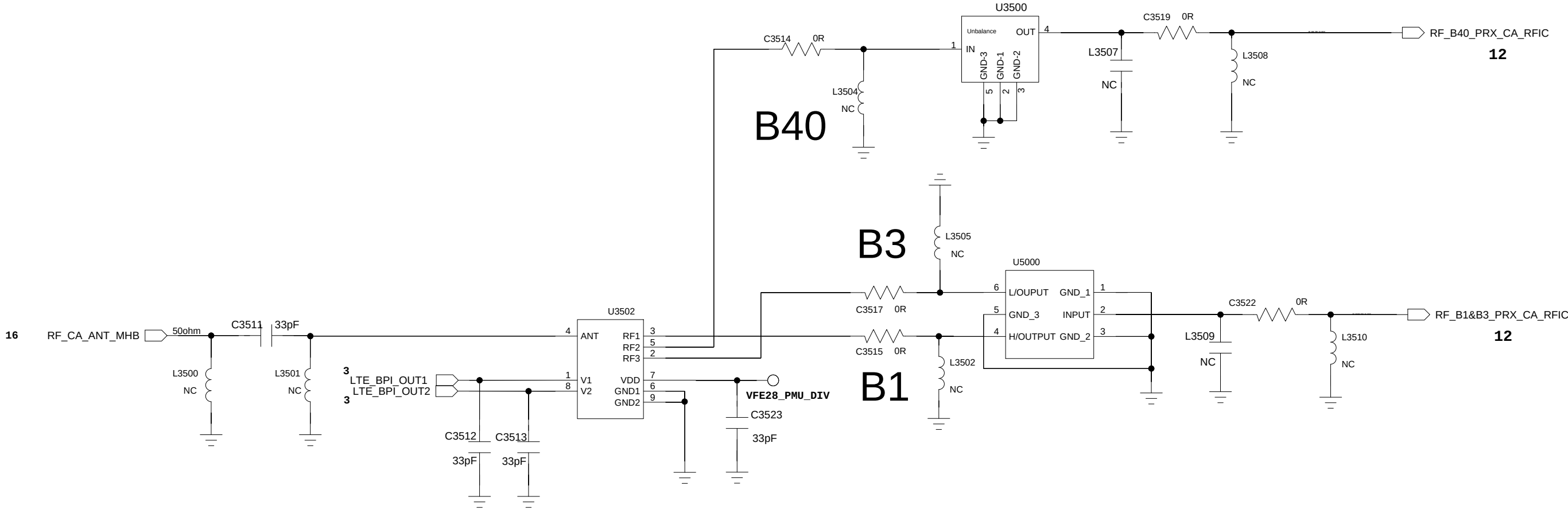
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LTR	ECO NO:	APPROVED:	DATE:



RF_MT6186M_RF_DRX

SP8T_LB

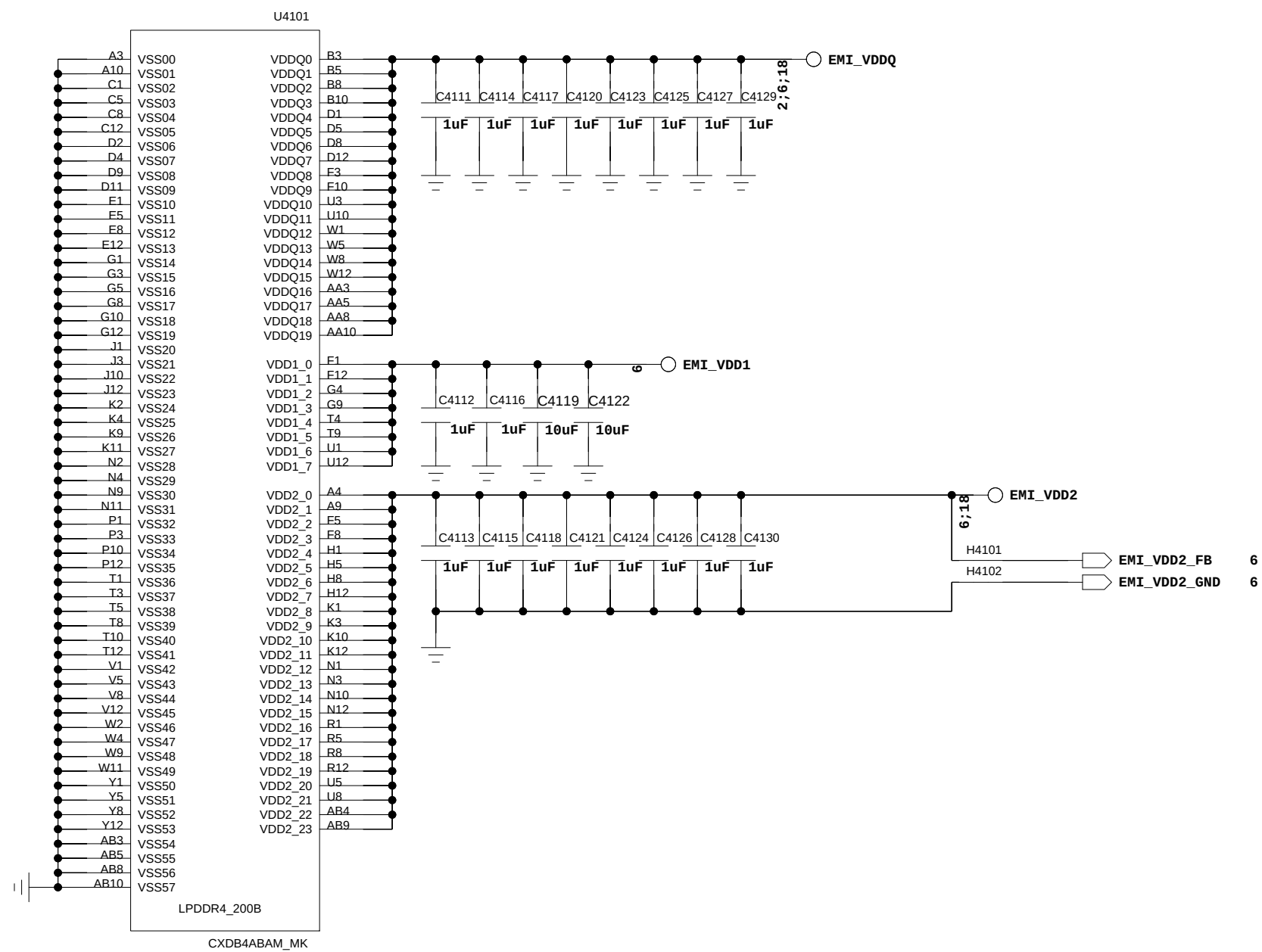
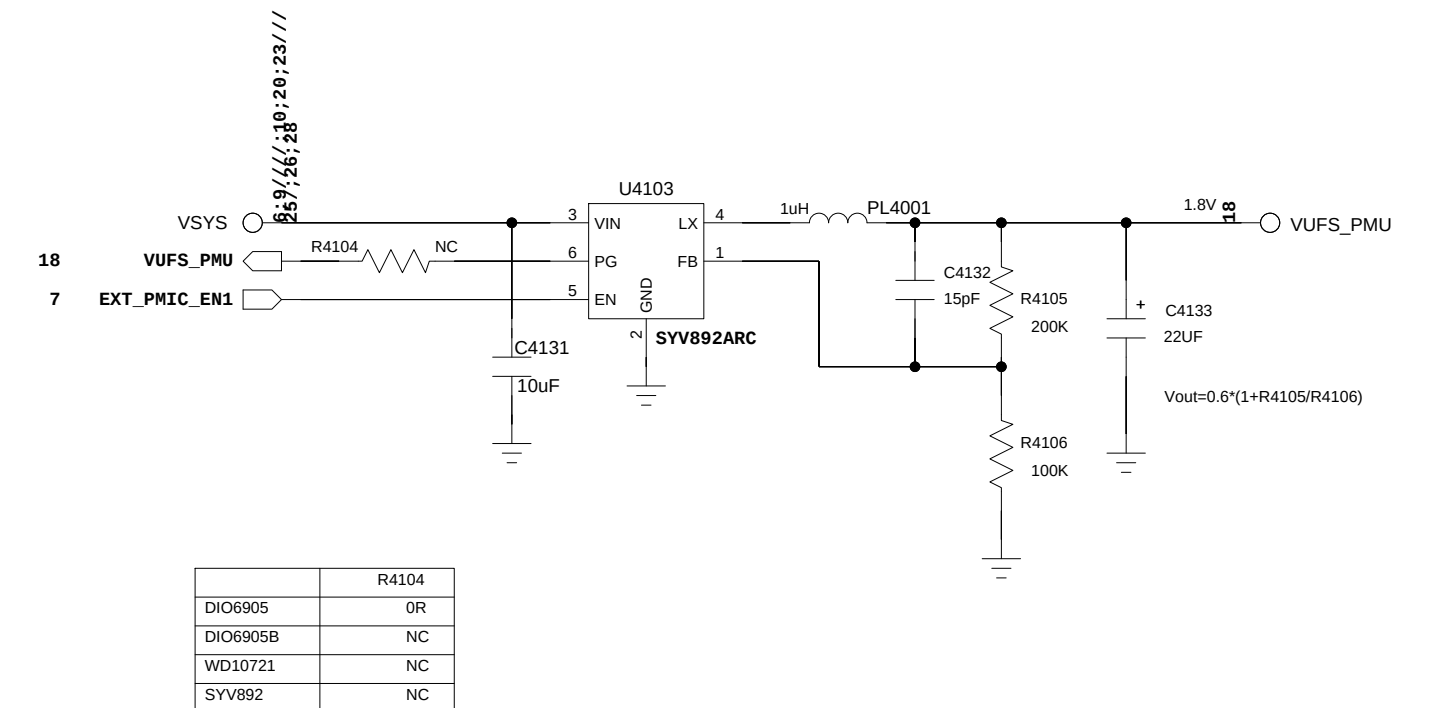
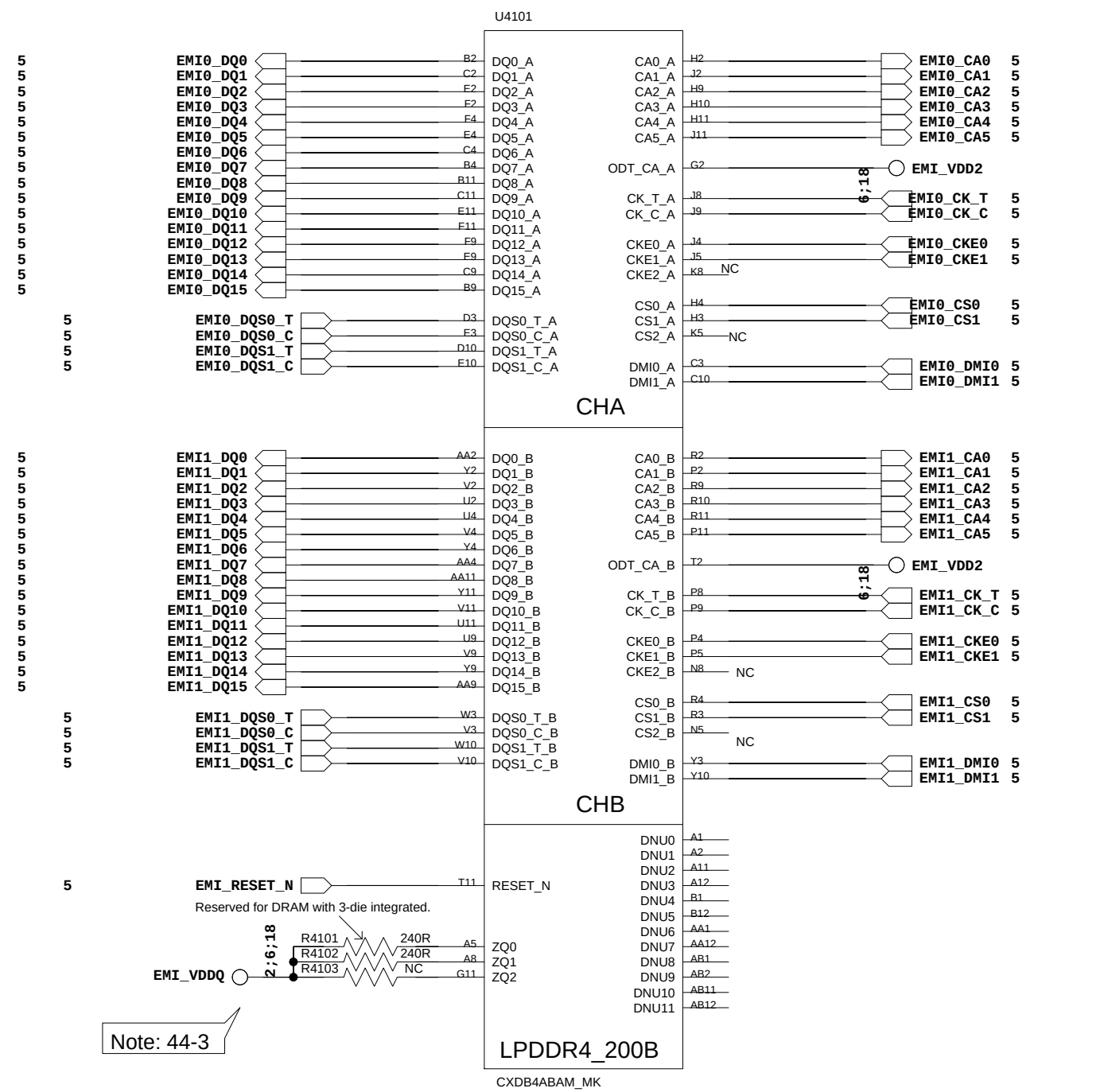
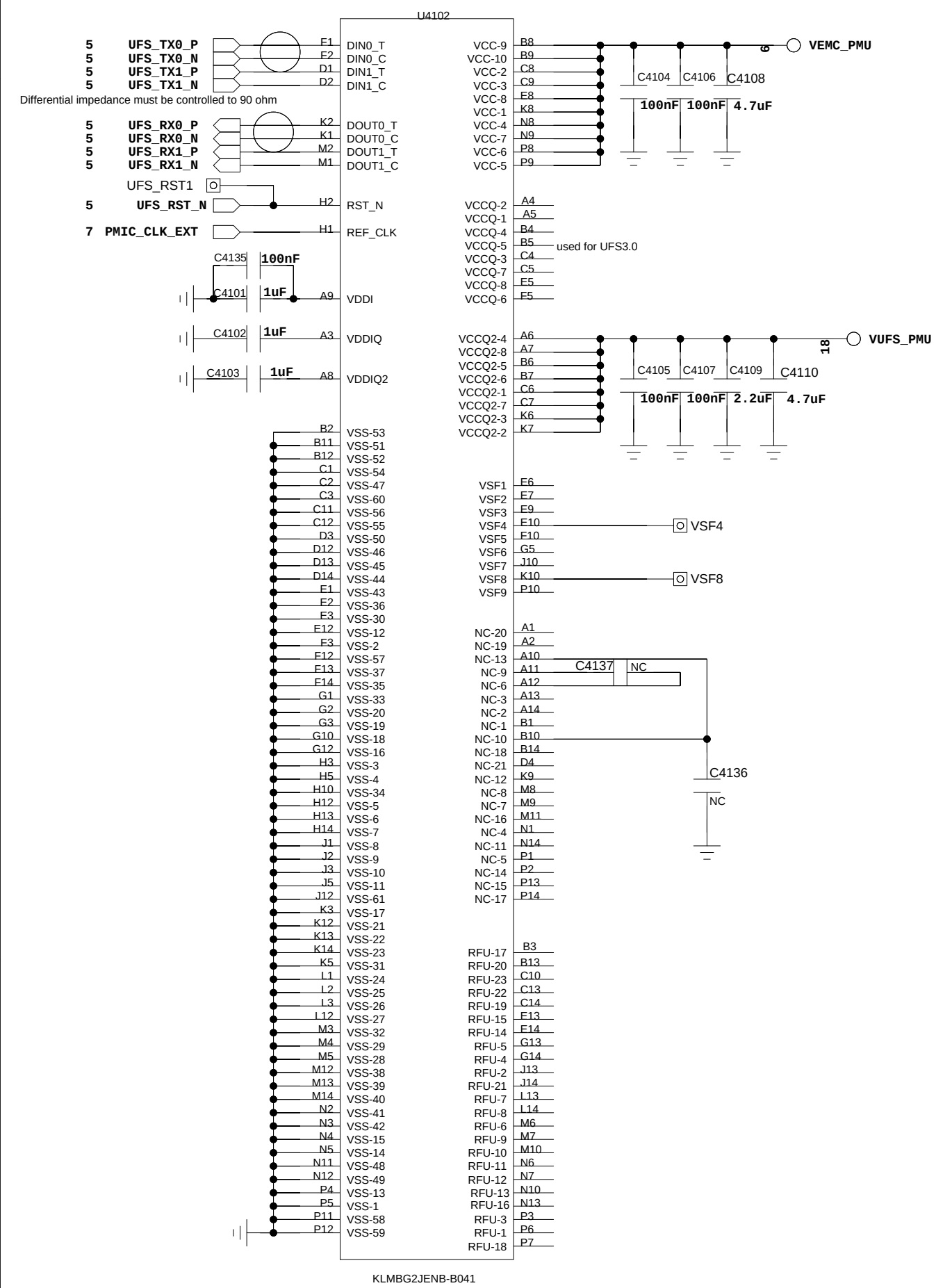
VC1613 control logic					SP8T control logic										REVISION RECORD			
VC1	VC2	RF1	RF2	RF3	SP6T control logic										LTR	ECO NO:	APPROVED:	DATE:
H	L	Y	N	N	V1	V2	V3	RF1	RF2	RF3	RF4	RF5	RF6	RF7	RF8			
H	H	N	Y	N	L	L	L	Y	N	N	N	N	N	N	N			
N	H	N	N	Y	L	L	H	N	Y	N	N	N	N	N	N			
					L	H	L	N	N	Y	N	N	N	N	N			
					L	H	H	N	N	N	Y	N	N	N	N			
					H	L	L	N	N	N	N	Y	N	N	N			
					H	L	H	N	N	N	N	N	Y	N	N			
					H	H	L	N	N	N	N	N	N	Y	N			
					H	H	H	N	N	N	N	N	N	N	Y			



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 27/06/2023:16:28	
DRAWN		DATED		TITLE: 35_RF_MT6186M_RF_MHB_CA		VERSION:	SHEET: 17 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

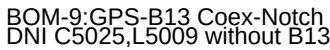
MEM_DSC_LPDDR4x

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 29/05/2023:16:22	
DRAWN		DATED		TITLE: 41_MEMORY_DSC_LPDDR4X		VERSION:	SHEET: 18 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

ANT12(GPS L1)£©



19 FM_AVDD28 0.2mm OR B5001 0.25mm VCN28_PMU 6;19

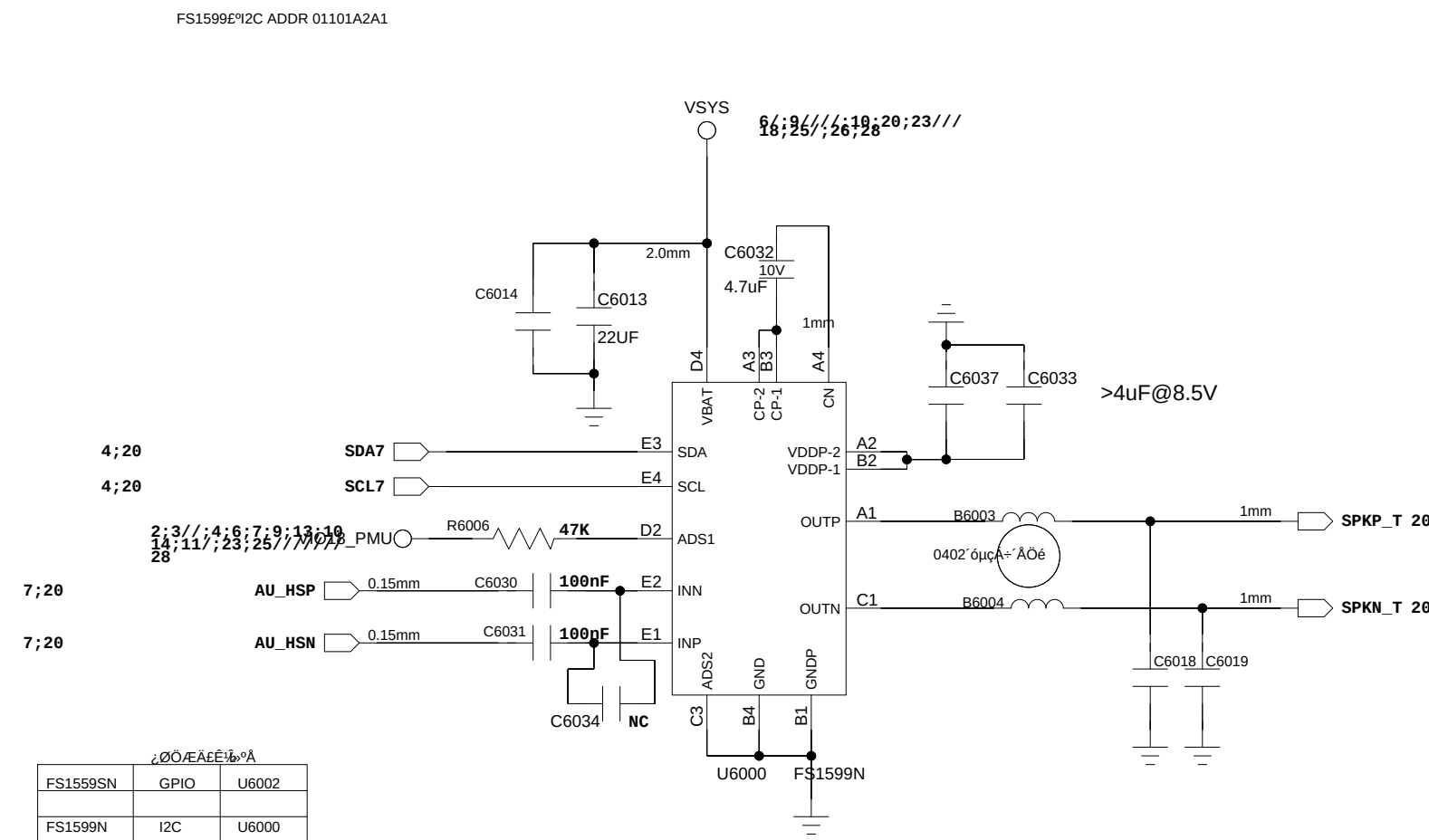
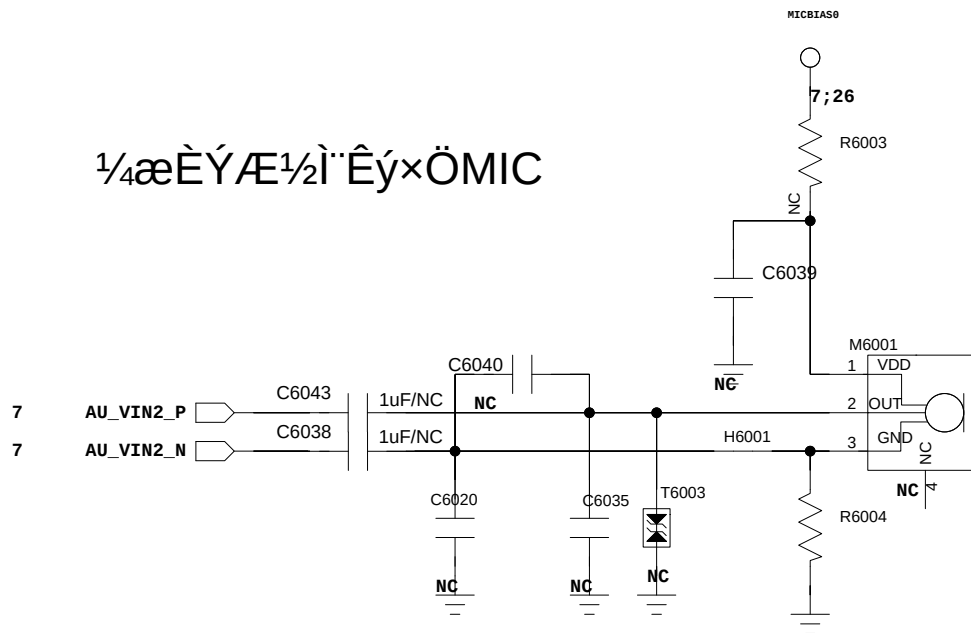
19 AVDD18_GPS 0.2mm H5003 Star connection 0.25mm VCN18_PMU 6

19 AVDD18_WBT 0.2mm H5004 0.25mm VCN18_PMU 6

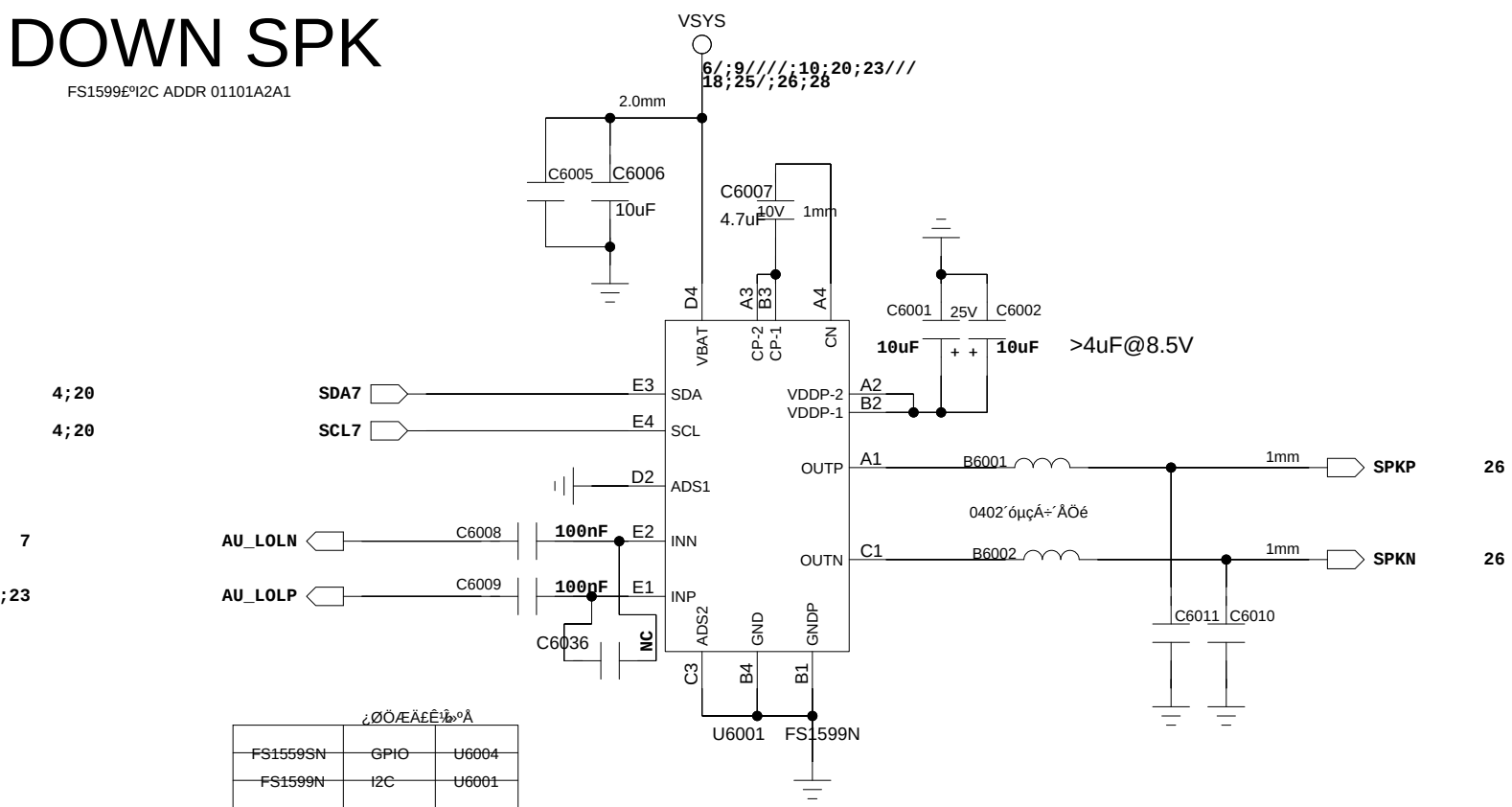
Note: For MT6631 need to configure the VCN33_2_PMU to 2.8V by PMIC SW.

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 19/06/2023:17:47	
DRAWN		DATED		TITLE: 50_CONNECTIVITY_MT6631		VERSION:	SHEET: 19 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

Audio PA
UP SPK


$$1/4 \text{æ} \text{È} \acute{Y} \text{Æ} 1/2 \grave{\text{I}} \text{¨} \text{Ê} \acute{y} \times \text{Ö} \text{MIC}$$
[illegible]

Audio PA
DOWN SPK



7 AU_VIN0_P C6041 1uF 0.1mm MICP0 26

7 AU_VIN0_N C6042 1uF 100pF 0.1mm MICN0 26

C6004 33pF

[illegible]

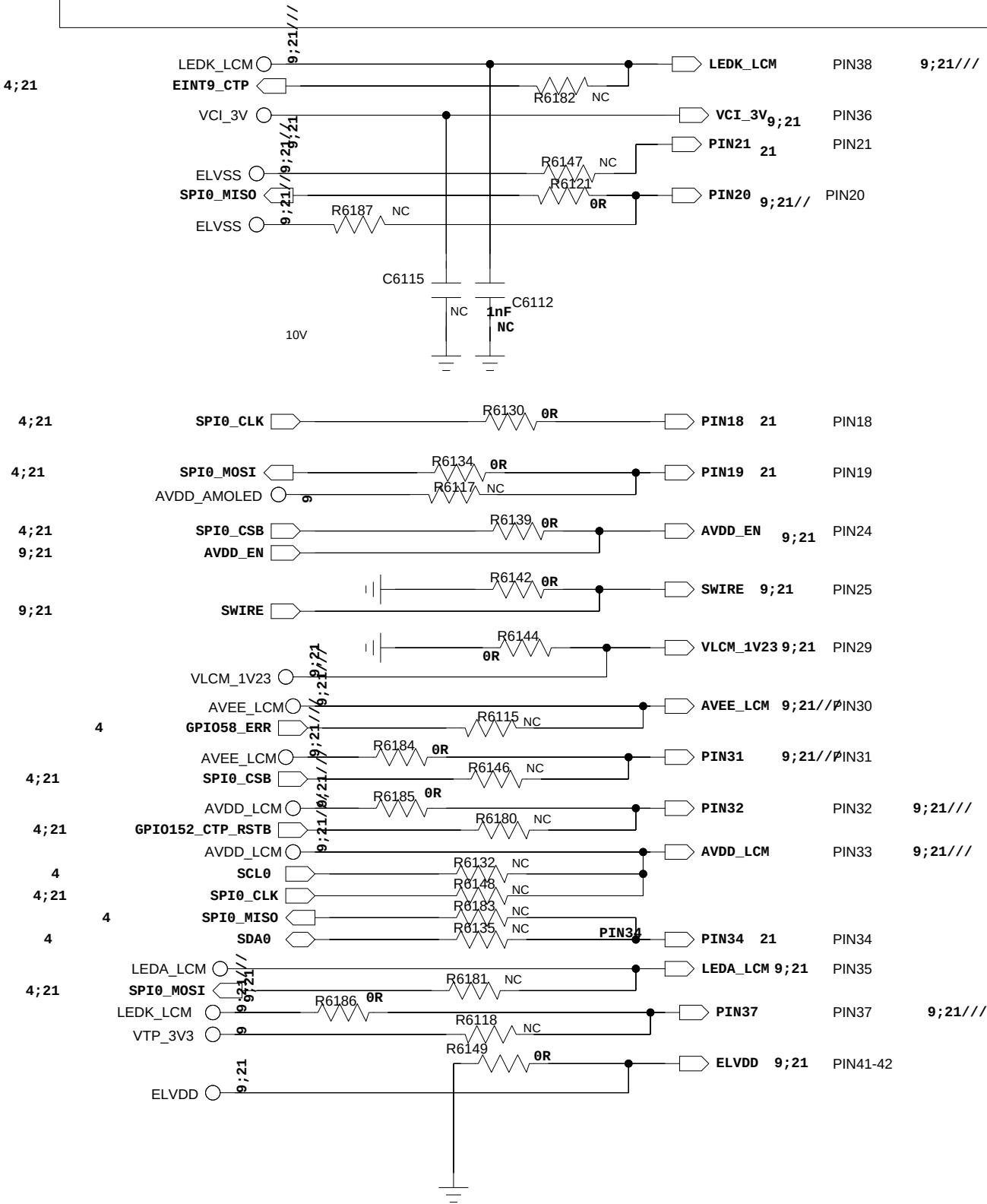
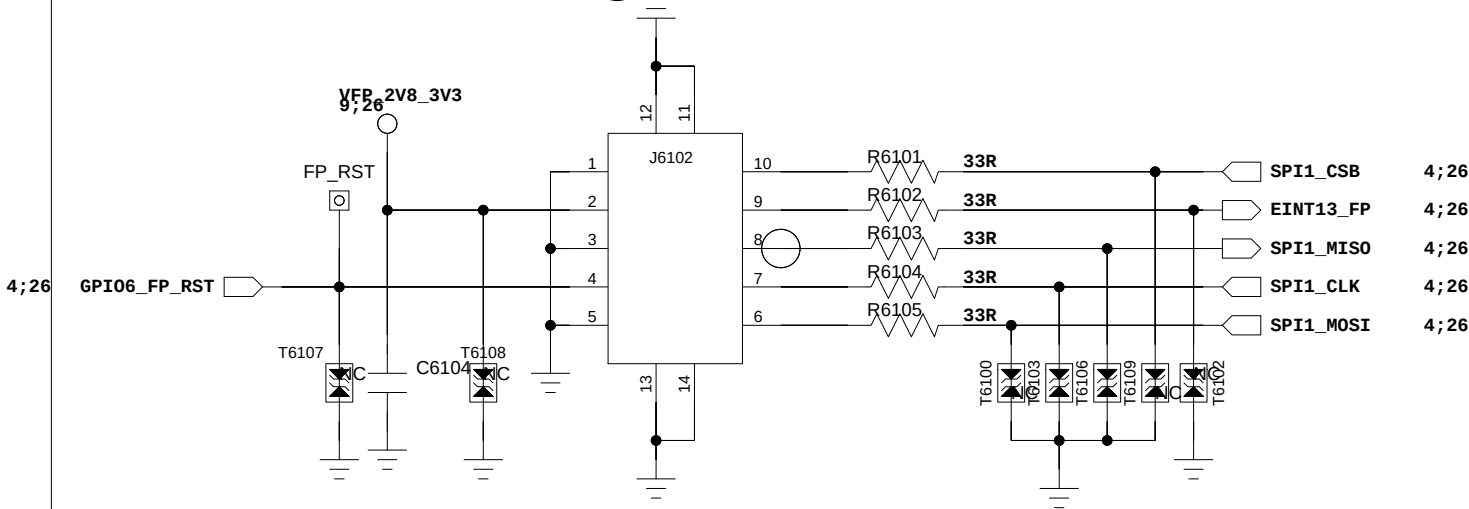
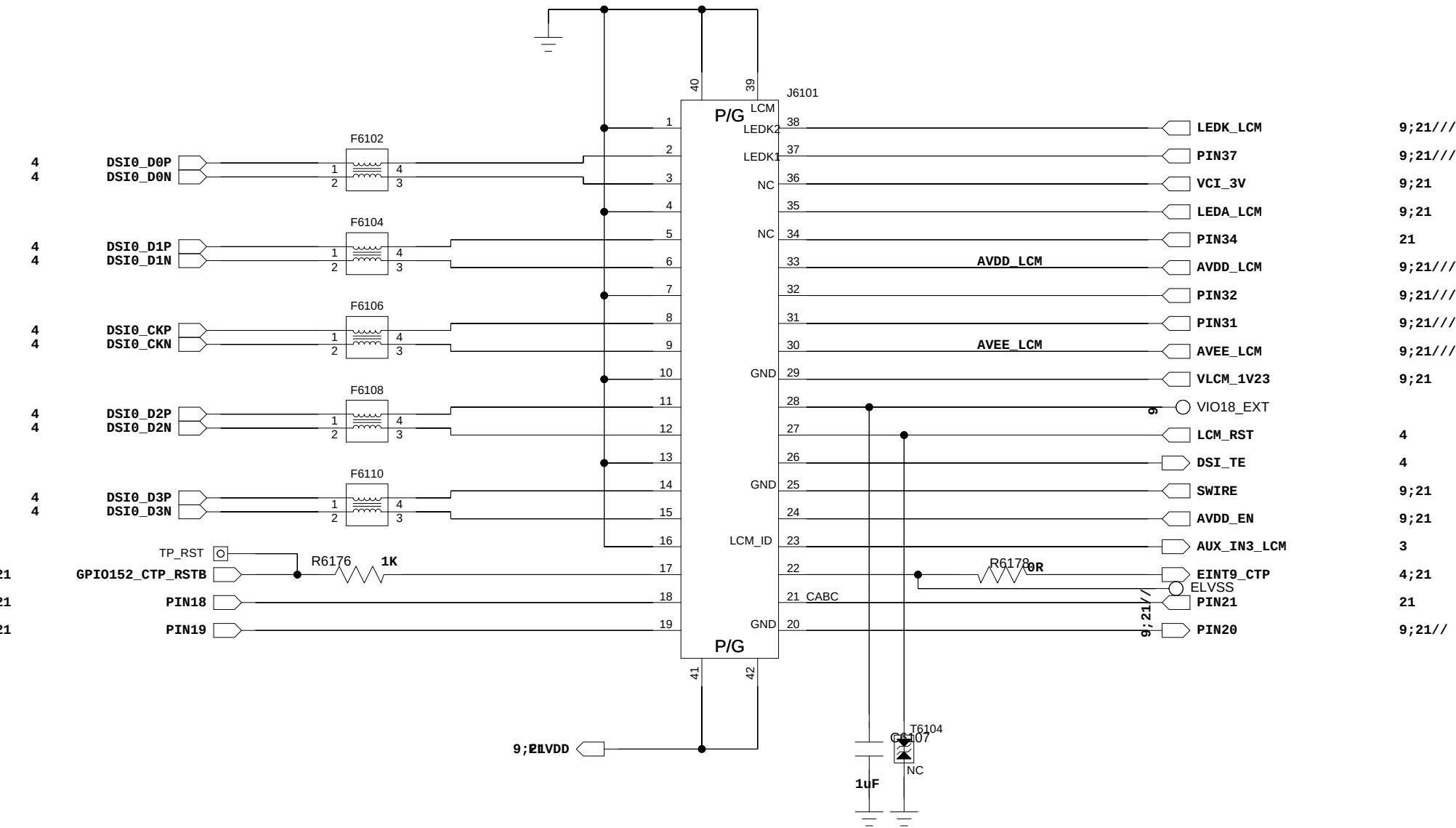
Based on your system level design , if better Audio performance is needed on your system, please add 320hm to audio path for performance enhance proposal (320hm condition pop noise can improve 6dB)

Schematic design notice of "60_PERI_AUDIO_IO" page.	
Note 60-1:	B6009 B6010 B6013 B6014 needs change to "BLM18BD102SN1" for high THD performance(-90dB), but this BOM change will result in FMS RSL10dB degraded.
Note 60-2:	To reserve a resistor in HPL and HPR in series connection both in order to optimize headphone pop noise. The recommended value of this resistor is 33R.
Note 60-3:	Layout trace from MT6353 ball J3 AUDREFIN to Audio jack GND must surround shield with GND.
Note 60-4:	0.1/uF for ACC mode(1uF for VB_ AMR Speech/0.1uF for NB_ AMR Speech),0R for DCC mode.

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 20/07/2023:21:16	
DRAWN		DATED		TITLE: 60_PERI_AUDIO		VERSION:	SHEET: 20 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

CTP (NT36525) I2C address: 0X01 (Write:0x02, Read:0x03)



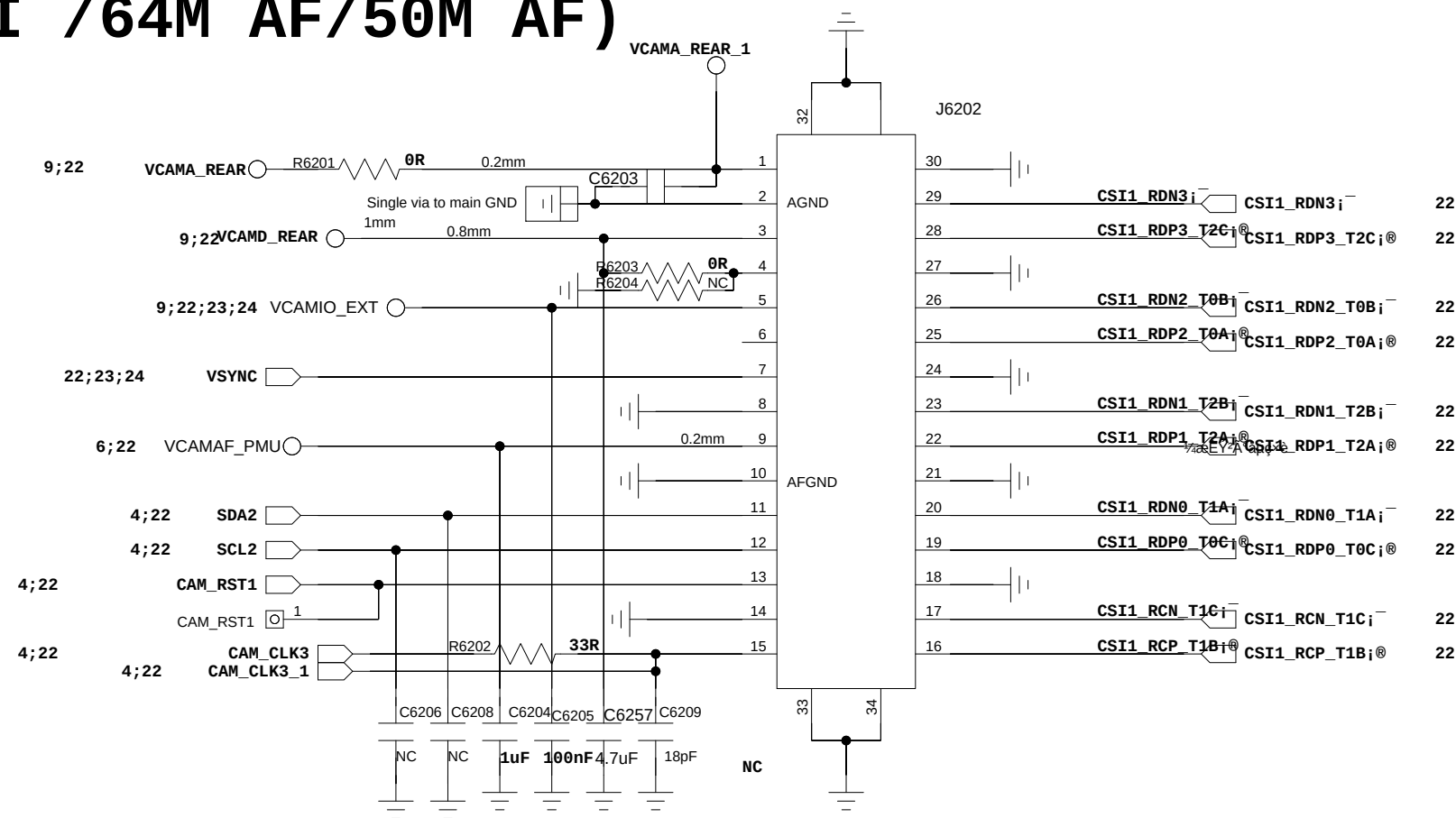
	OLED	
LCD	TP:12C	TP:SPI
R6121	R6115	R6115
R6130	R6117	R6117
R6134	R6118	R6118
R6139	R6132	R6148
R6142	R6135	R6183
R6144	R6146	R6146
R6149	R6147	R6147
R6176	R6180	R6180
R6178		R6181
R6184	R6182	R6182
R6185	R6187	R6187
R6186		

LCD

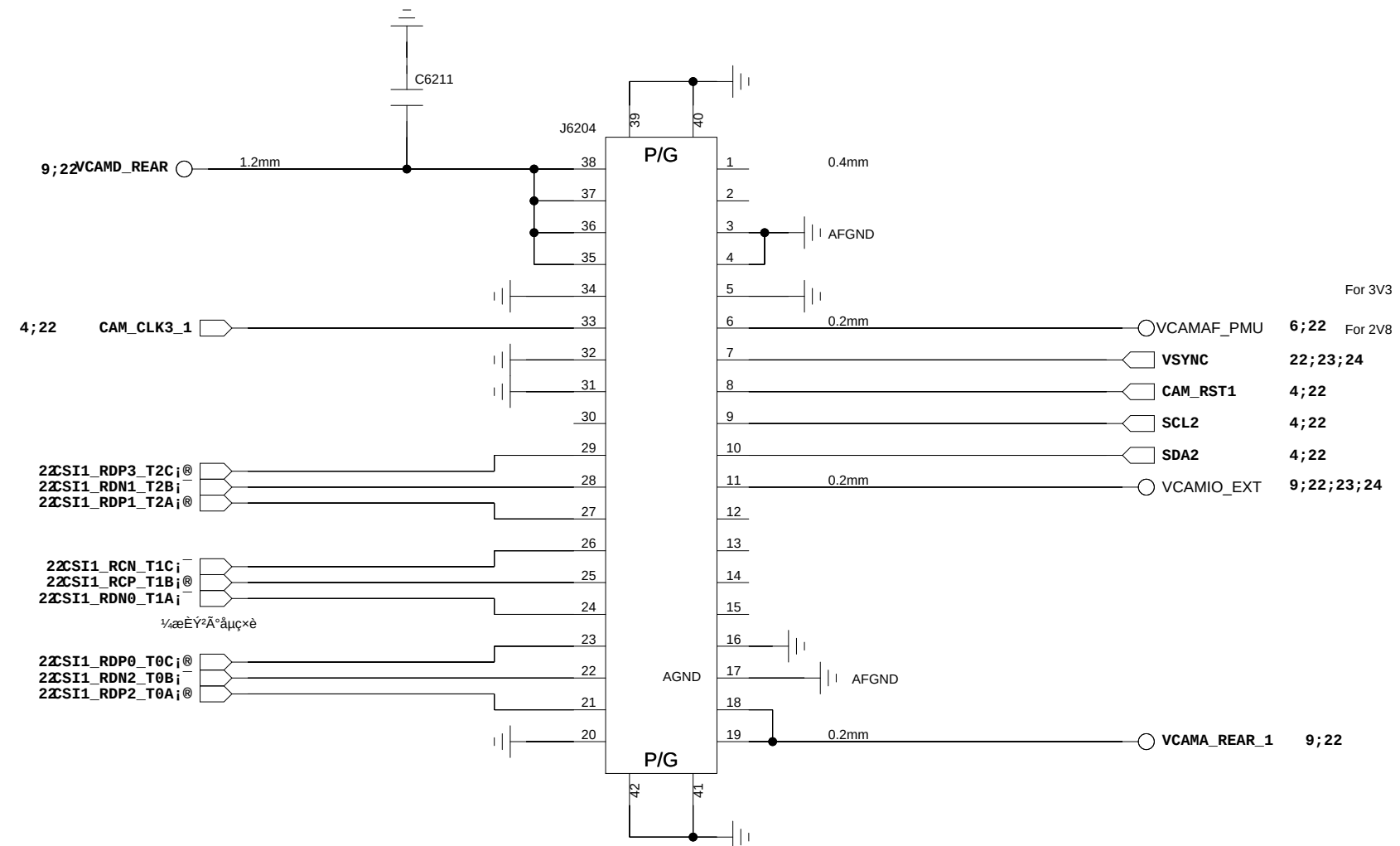
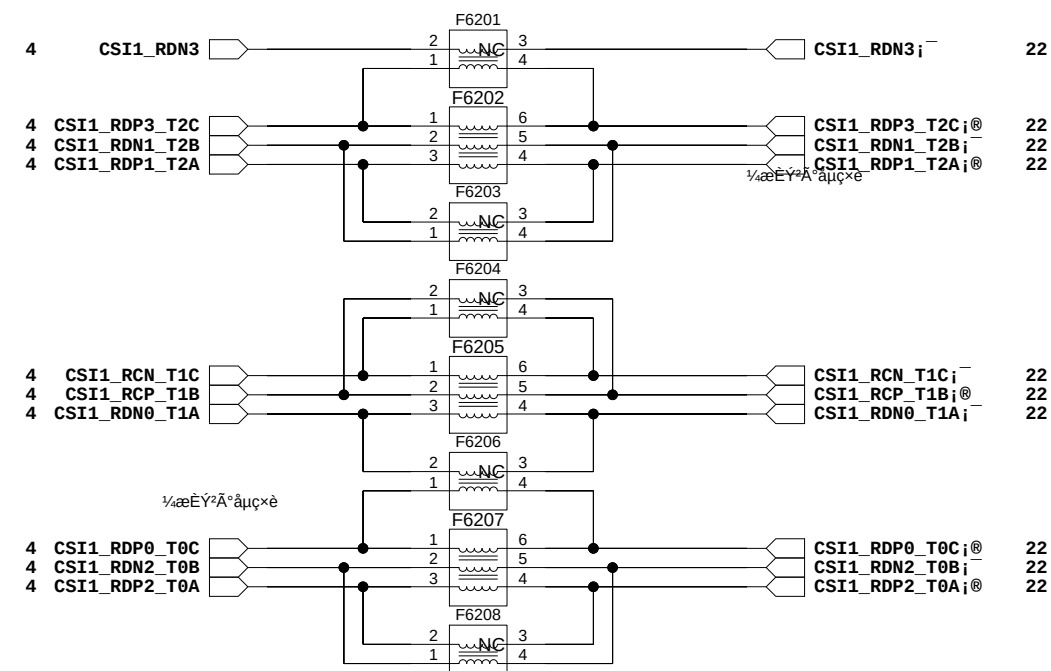
OLED

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 01/06/2023:17:31	
DRAWN		DATED		TITLE: 61_PERI_LCM_CTP_FP		VERSION:	SHEET: 21 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_CAMERA_I
(CAM I /64M AF/50M AF)



	AVDD	DVDD	IOVDD	AFVDD	I2C addr
50M AF-17201711 50M AF-17201715 55KJN1SQ03-F6X9	2.8V 67.7mA	1.05V 260.7mA	1.8V 4mA	2.8V 109mA 2.8V 103mA	Sensor: 0x20(W)/0x21(R) VCM: 0x18(W)/0x19(R) EEPROM: 0xA0(W)/0xA1(R)
50M AF Hi-5021Q	2.8V 78mA	1.1V 598mA	1.8V 31mA	2.8V 103mA	Sensor: 0x40(W)/0x41(R) VCM: 0x18(W)/0x19(R) EEPROM: 0xA0(W)/0xA1(R)
64M AF OV64B40	2.8V 89mA	1.1V 600mA	1.8V 1.8mA	2.8V 105mA	Sensor: 0x20(W)/0x21(R) VCM: 0x18(W)/0x19(R) EEPROM: 0xA0(W)/0xA1(R)



	AVDD	DVDD	IOVDD	AFVDD	OIS-VDD	I2C addr
108M S5KHM6SX03-FGX9	2.2V120mA	1.0V1200mA	1.8V50mA	3.3V100mA	3.3V540mA	Sensor: 0x?? (W) / 0x?? (R) VCM: 0x?? (W) / 0x?? (R) EEPROM: 0x?? (W) / 0x?? (R)
64M KGWBSW03-FGX9	2.8V 163.0mA	1.0V 750mA	1.8V 18.55mA	2.8V 105mA	3.3V540mA	Sensor: 0x20 (W) / 0x21 (R) VCM: 0x18 (W) / 0x19 (R) EEPROM: 0xA0 (W) / 0xA1 (R)
108M S5KHM6SX03-FGX9	2.2V 88.6mA	1.1V 726.3mA	1.8V 2.25mA	2.8V 105mA	3.3V540mA	Sensor: 0x20 (W) / 0x21 (R) VCM: 0x18 (W) / 0x19 (R) EEPROM: 0xA0 (W) / 0xA1 (R)
50M OV50A40	2.8V 96mA	1.1V 780mA	1.8V 0.1mA	3.3V 100mA	3.3V540mA	Sensor: 0x20 (W) / 0x21 (R) VCM: 0x18 (W) / 0x19 (R) EEPROM: 0xA0 (W) / 0xA1 (R)
50M OV50A40	2.8V 96mA	1.1V 780mA	1.8V 2.1mA	2.8V 105mA	3.3V540mA	Sensor: 0x20 (W) / 0x21 (R) VCM: 0x18 (W) / 0x19 (R) EEPROM: 0xA0 (W) / 0xA1 (R)

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 07/06/2023:21:19	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_I		VERSION:	SHEET: 22 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

CAM II (2M)

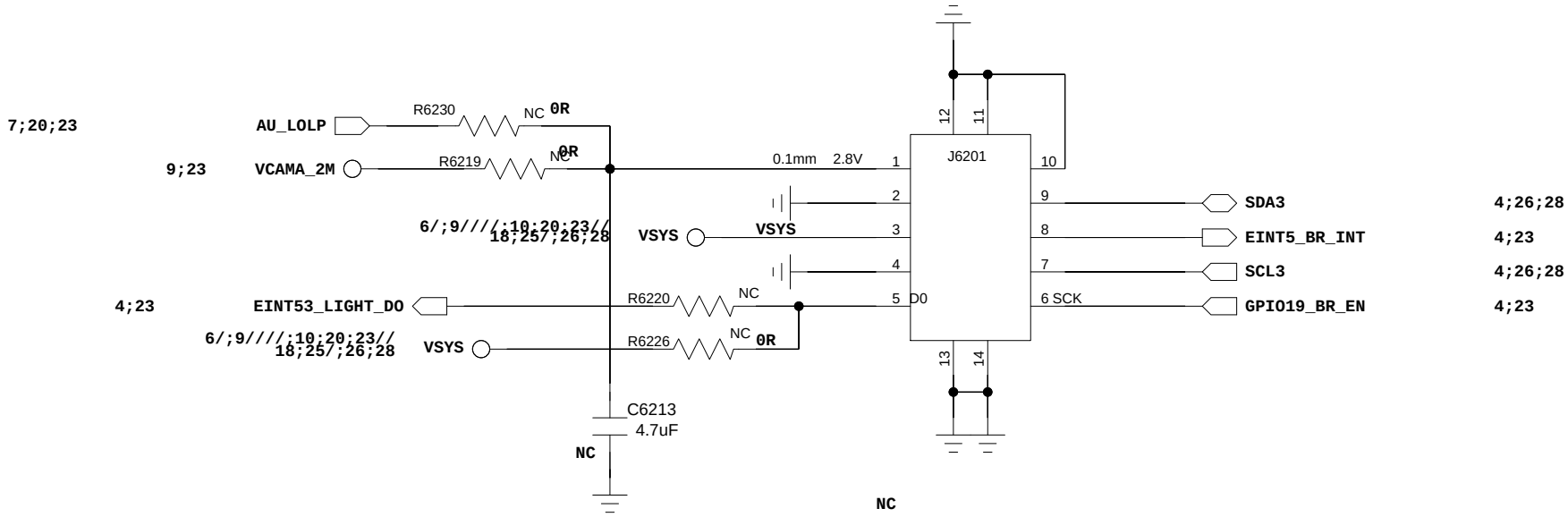
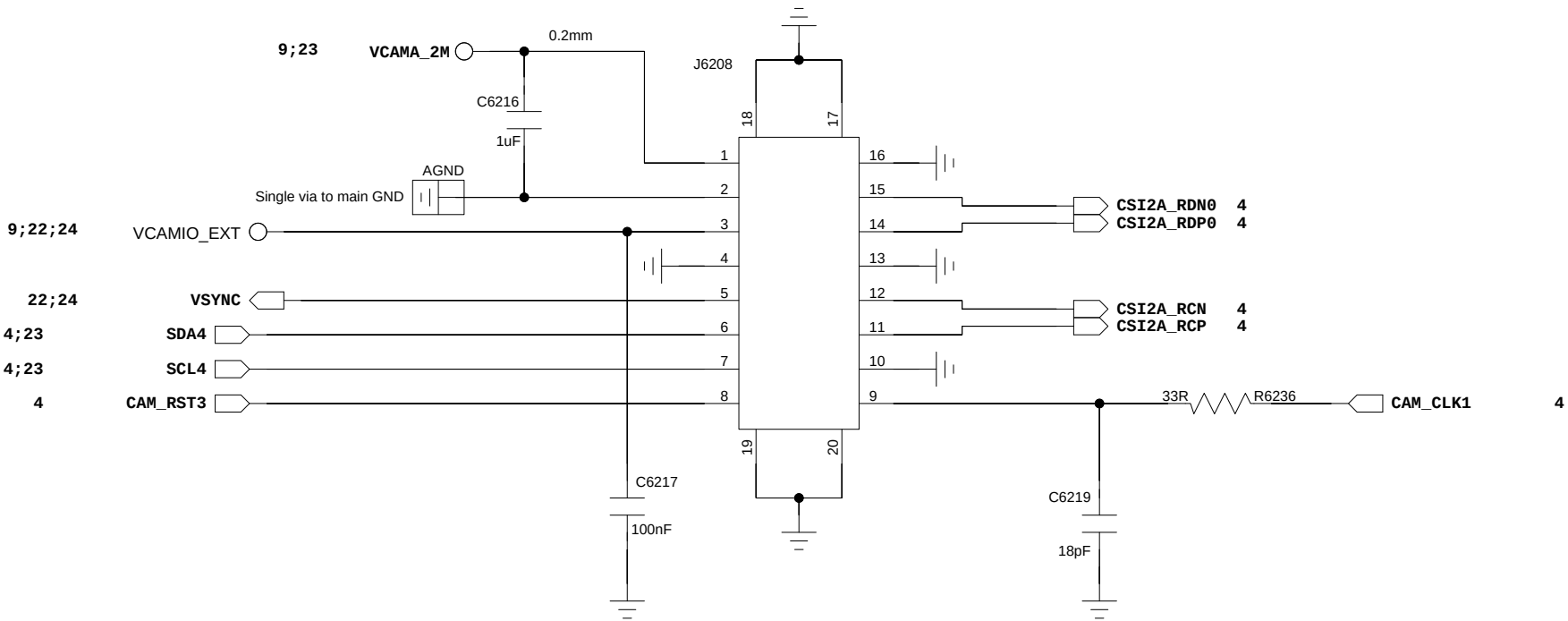
17201755 GC02M1B-C24Y0; I2C addr:0x20(W)/0x21(R)

17201679 GC02M1B-C24Y0; I2C addr:0x20(W)/0x21(R)

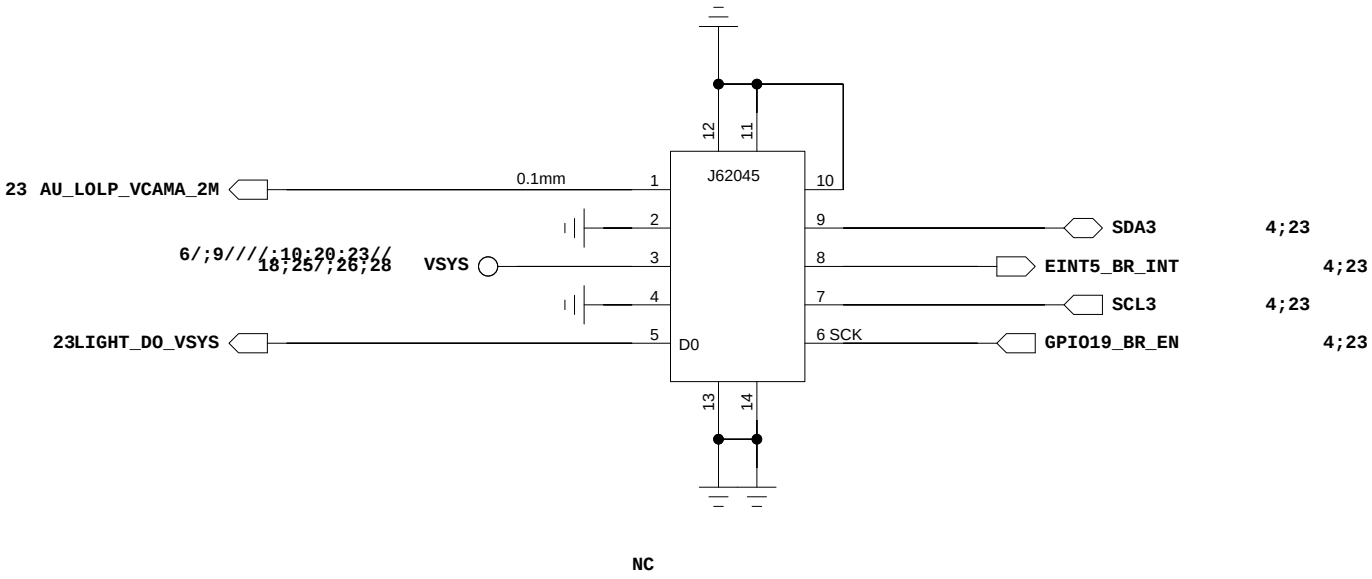
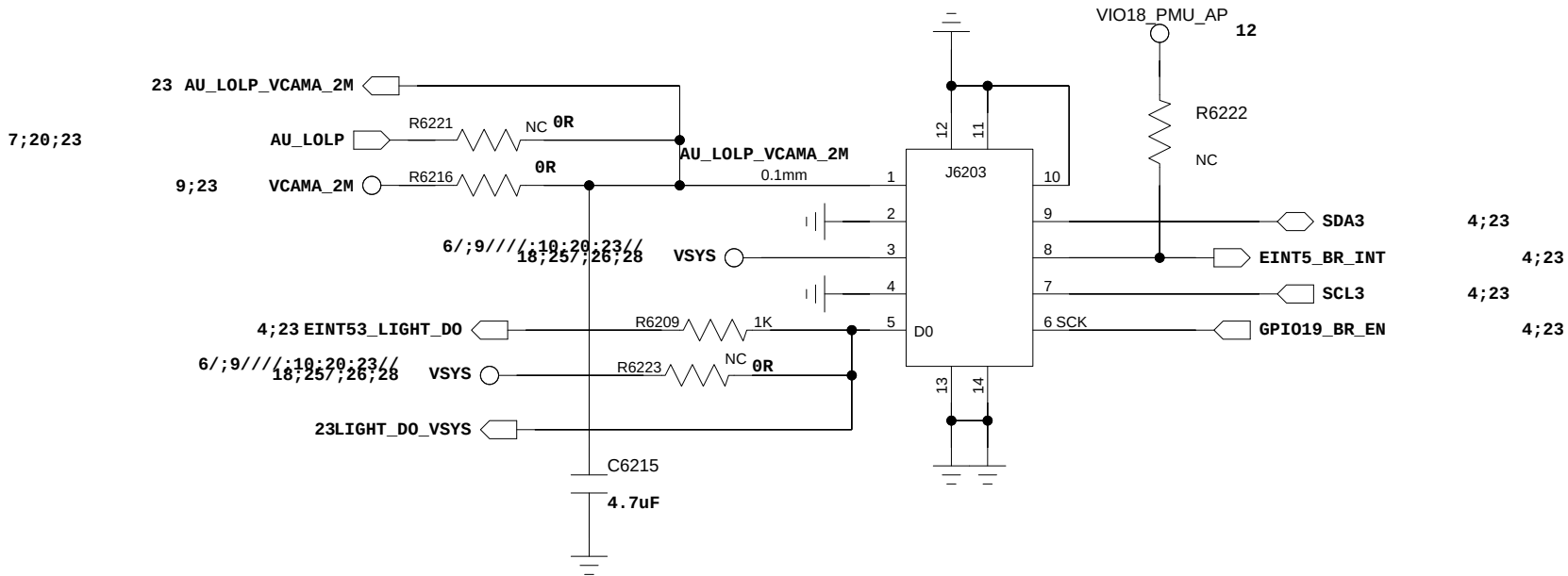
CAM III (Q)

17201679 GC02M1B-C24Y0; I2C addr:0x20(W)/0x21(R)

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LTR	ECO NO:	APPROVED:	DATE:



CAM II (Q+)



PIN1	2V8	VSYS
PIN5	DO	RGB_INT
PIN9	SDA	SDA
PIN8	SCL	SCL
PIN7	CAM_RST	RGB_EN
PIN6	NC	Audio_IN

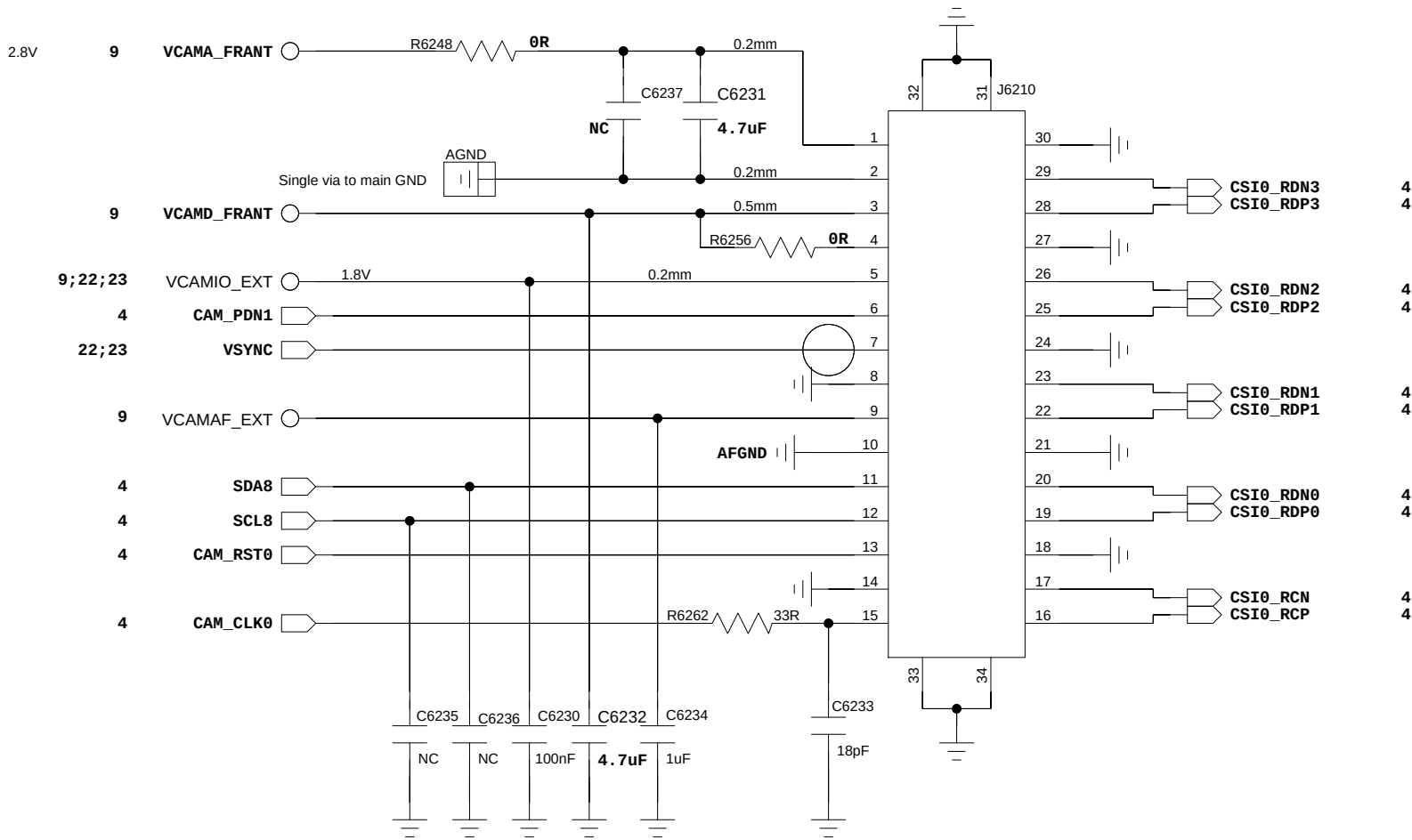
	AVDD	DVDD	IOVDD	I2C addr
2M(17201755) GC02M1-C24Y0 (CSP)	2.8V,40mA	/	1.8V,70mA	sensor:0x20(W)/0x21(R)
2M(17201677) GC02M1-C24YA (CSP)	2.8V,40mA	/	1.8V,70mA	sensor:0x20(W)/0x21(R)
2M(17201678) OV02B10	2.8V,33mA	/	1.8V,34mA	sensor:0x78(W)/0x79(R)

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 07/06/2023:22:56	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_II		VERSION:	SHEET: 23 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_CAMERA_III

FRONT MAIN CAMERA (16M/32M/50M AF)

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



		AVDD	DVDD	IOVDD	AF	4Pin	I2C addr
16M	17201705 OV16810	2.8V52mA	1.05V200mA	1.8V5mA		DVDD	Sensor: 0x6C(W)/0x6D(R) EEPROM: 0xA2(W)/0xA3(R)
32M	17201766 S5KGD2SP03-F6X9	2.8V55.2mA1.0V 468mA	1.8V4mA			DVDD	Sensor: 0x5A(W)/0x5B(R) EEPROM: 0xA2(W)/0xA3(R)
50M	S5KJN1	2.8V67.7mA1.1V260.7mA	1.8V4mA	2.8V100mA		DVDD	Sensor: 0xAC(W)/0xAD(R) EEPROM: 0xA2(W)/0xA3(R) AF: 0X1C(W)/0X1D(R)

COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 07/06/2023:21:17	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_III		VERSION:	SHEET: 24 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

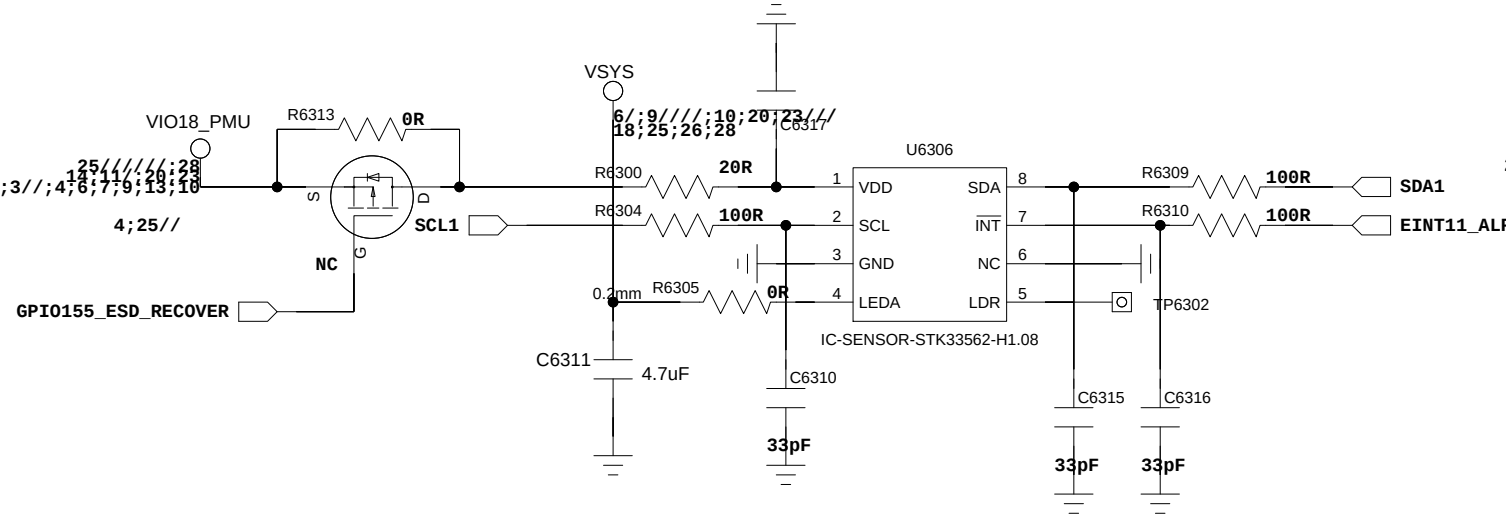
PERI_SENSORS

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

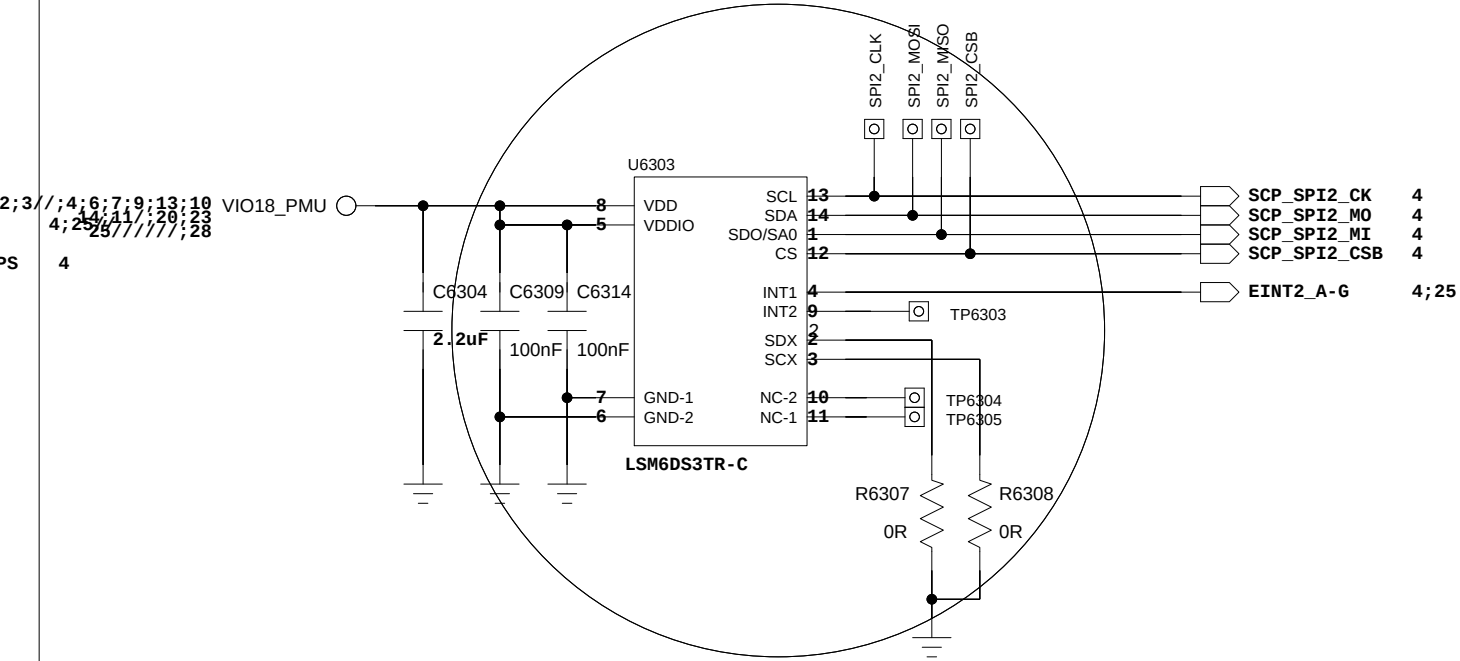
AL& PS Sensor

STK33562:I2C address: Write:0x8C, Read:0x8D
LTR-569ALS-WA:I2C address: Write:0x46, Read:0x47
MN78911D:I2C address: Write:0x82, Read:0x83

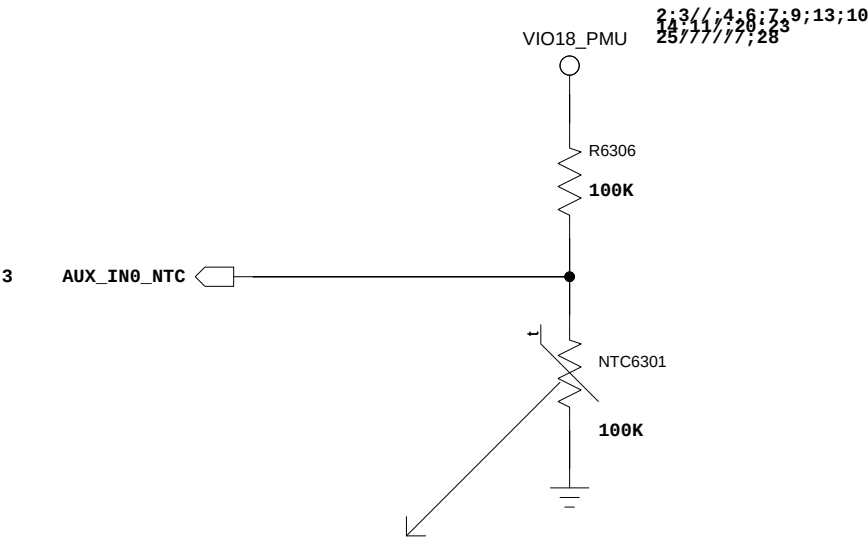
Change to STK3335X



G-Sensor + Gyro Sensor



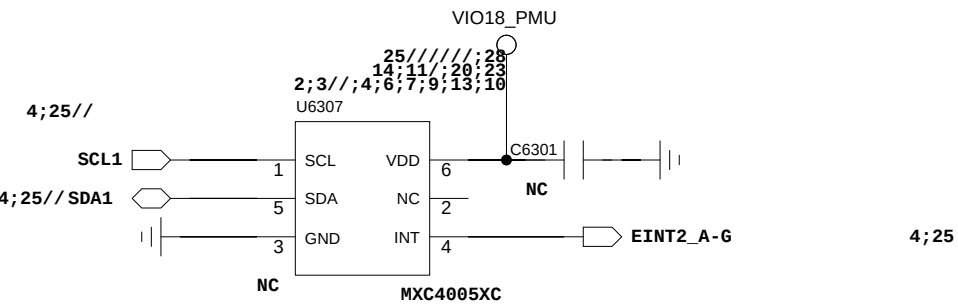
Thermistor to sense AP temperature



- 1. NTC6301must keep a distance about 6-8 mm away from BB and far from other heat sources 10 mm at least.
- 2. The distance is the shortest distance from package edge to edge.

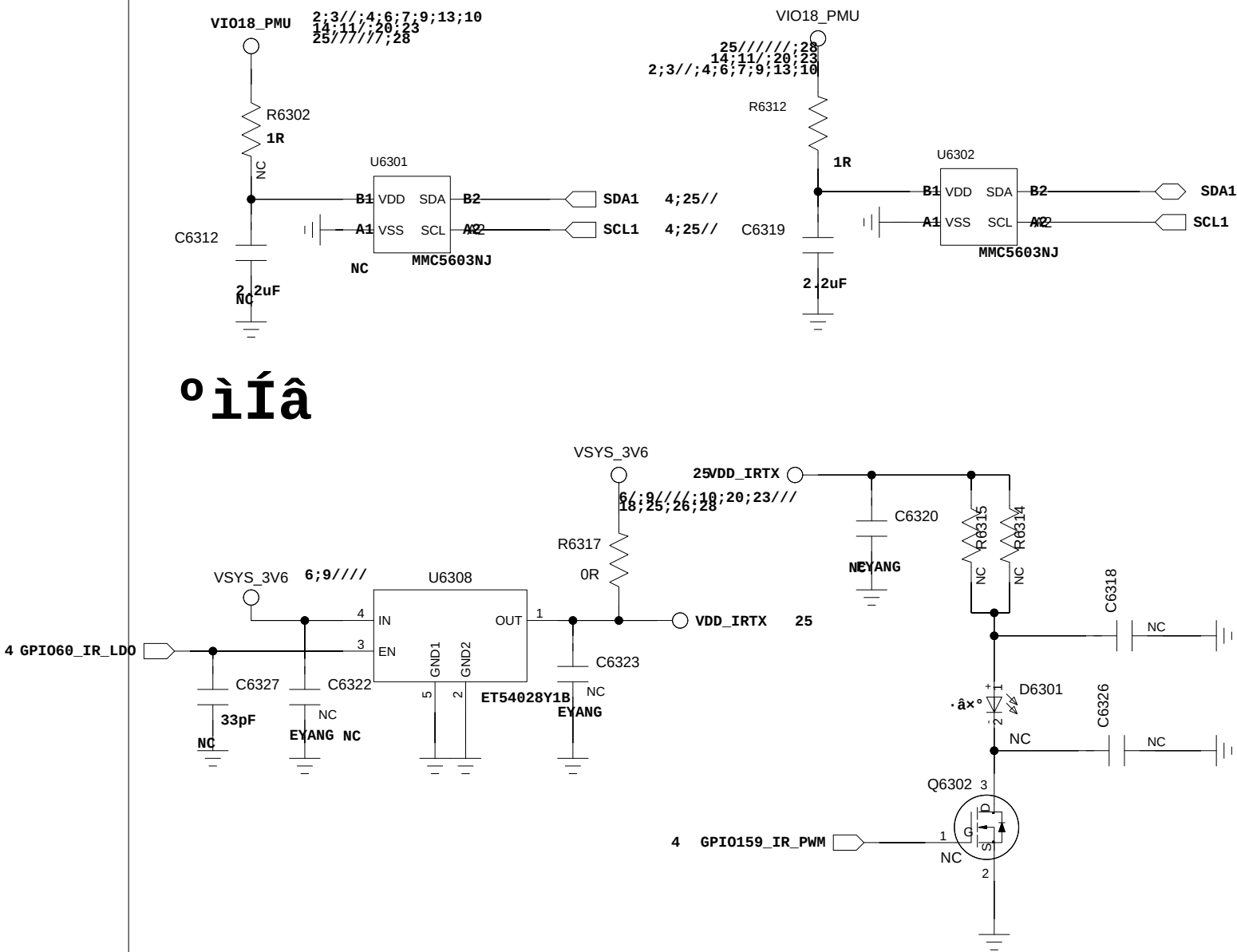
G-Sensor

MXC6655XA:ADDRESS:0x2A(Write)/0x2B(Read)
SC7A20:ADDRESS:0x30(Write)/0x31(Read)
MXC4005XC:ADDRESS:0x2A(Write)/0x2B(Read)



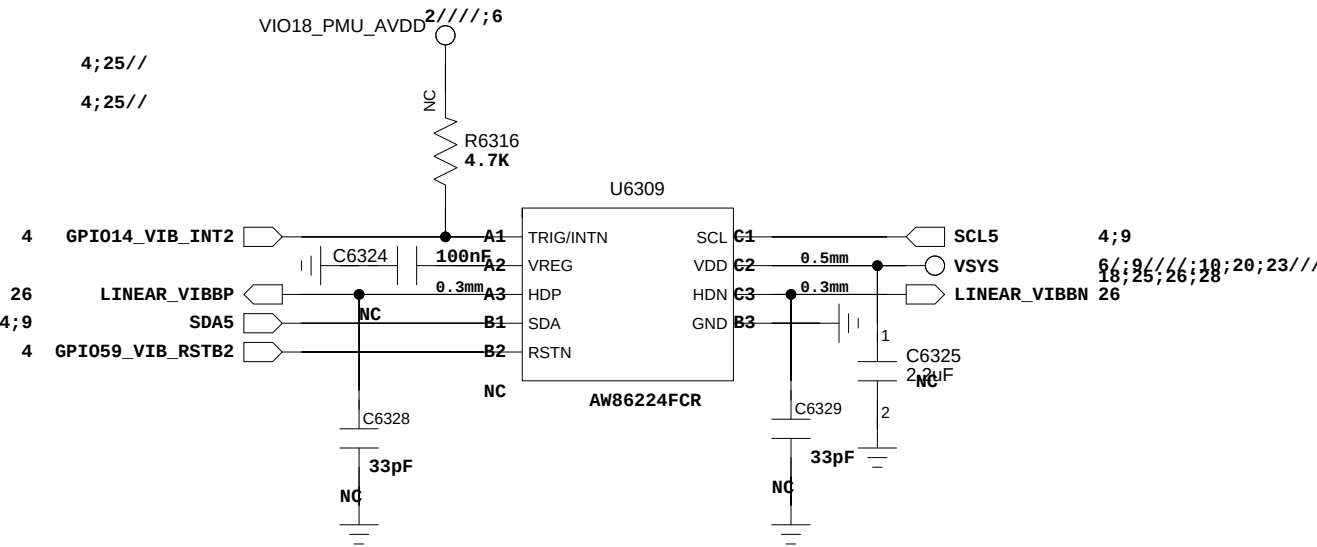
M-Sensor(COMPASS)

MMC5603:I2C ADDRESS:0x60(Write)/0x61(Read)
QMC6308:I2C ADDRESS:0x58(Write)/0x59(Read)



VIB Z Linear Motor Driver

AW86224FCR:I2C ADDRESS:0xB0(Write)/0xB1(Read)



component	MXC6655XA	SC7A20	STK8BA50-S
C6302	NC	100nF	1uF
C6303	1uF	100nF	1uF

COMPANY: TRANSSION HOLDINGS

MODEL:G99_STD_MAIN

Modified Date: 17/06/2023:14:19

DRAWN

DATED

TITLE: 63_PERI_SENSORS

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DATED

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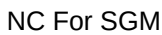
VERSION:

SHEET: 25 OF 28

Exchange IO2



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RT1715:I2C ADDRESS:0x9C(Write)/0x9D(Read)



4

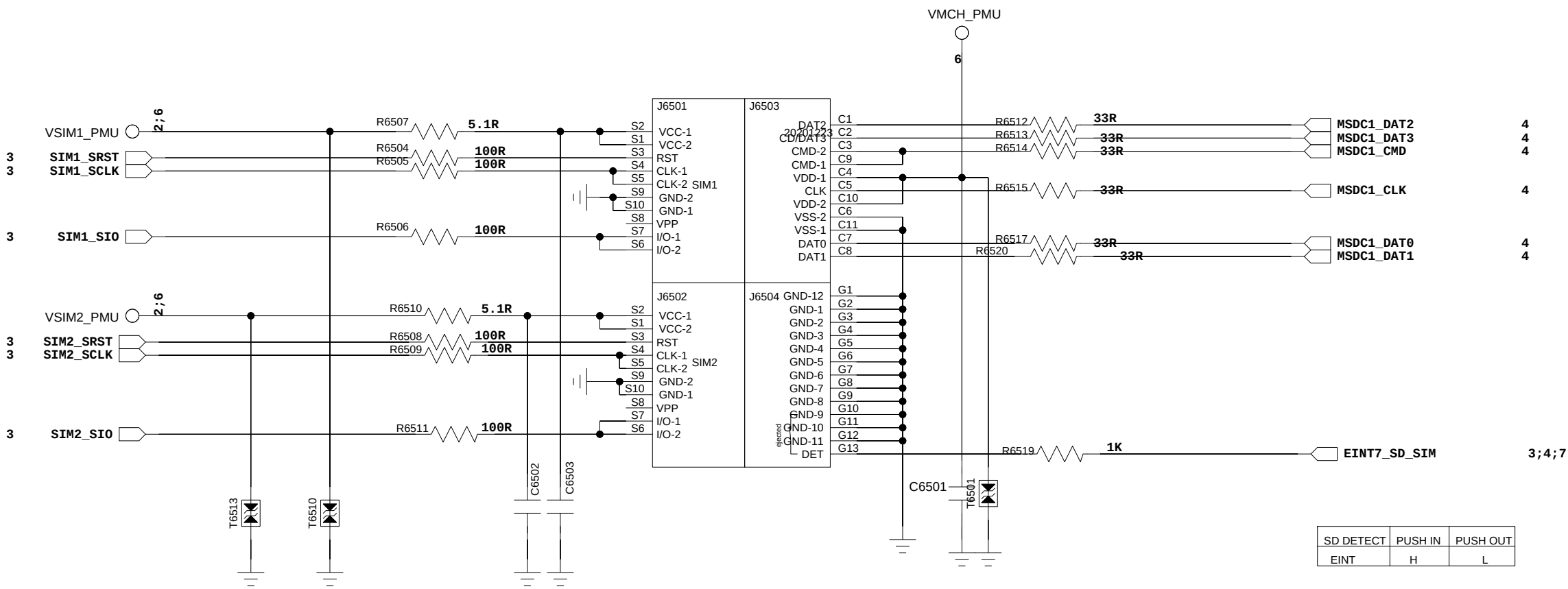
SC8546
80W
I2C5/1â¹0μçE½×ª»»

COMPANY: TRANSSION HOLDINGS				MODEL: G99_STD_MAIN		Modified Date: 12/06/2023:15:12	
DRAWN		DATED		TITLE: 64_PERI_EXCH_IO		VERSION:	SHEET: 26 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

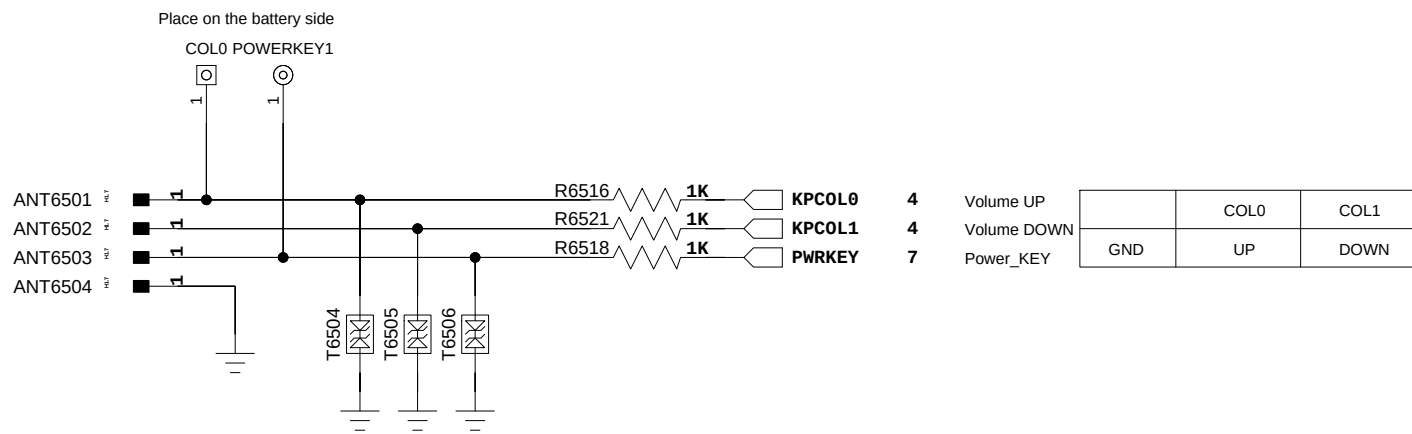
PERI_SIM_SD_KEYPAD

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

SIM1/2-SD-CARD



SIDEKEY



COMPANY: TRANSSION HOLDINGS				MODEL:G99_STD_MAIN		Modified Date: 29/05/2023:19:53	
DRAWN		DATED		TITLE: 65_SD_SIM_KEY		VERSION:	SHEET: 27 OF 28
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

